The Strategic Formation of Networks: Experimental Evidence *

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Abstract

We use a laboratory experiment to explore dynamic network formation in a six-player game where link creation requires mutual consent. The analysis of network outcomes suggests that the process converges to the pairwise-stable (PWS) network when a PWS network exists and not to converge to any network when a PWS network does not exist. When multiple Pareto-rankable PWS networks exist, subjects coordinate on the high-payoff one. The analysis at the single choice level indicates that the percentage of myopically rational behavior is high. Deviations from myopic rationality are more prevalent in early turns, when marginal payoff losses are small and when deviations involve excessive links that may be removed unilaterally later on.

Keywords: social networks, pairwise stability, myopic rationality, laboratory experiments.

JEL Classification: C73, C92, D85.

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1 Introduction

Social networks facilitate information exchange when formal channels are unavailable (Munshi, 2007). They also work as a means to smooth consumption in the absence of formal insurance markets (Udry, 1994; Fafchamps and Lund, 2003). Even markets can be seen not as anonymous institutions of exchange but as networks that facilitate exchange between buyers and sellers (Kranton and Minehart, 2001). Sociologists have, for a while now, recognized the importance of networks and social structures, and more recently, economists too have joined the fray (Granovetter, 2005; Jackson, 2005).

Within the economic literature, one question of interest is how networks are formed. Two seminal theoretical papers advanced explorations of this question for different link formation rules. Bala and Goyal (2000) study efficient and Nash-stable networks in non-cooperative games where links are made and broken unilaterally. Jackson and Wolinsky (1996) address a similar problem with an underlying game that requires links between two agents to be mutually agreed, although they can be broken unilaterally.

In addition, there is a small but growing experimental literature that builds on these theories.[1] However, the bulk of this literature focuses on the former type of rule, namely when links can be made and broken unilaterally (see e.g. the seminal paper by Callander and Plott (2005) as well as Berninghaus, Ehrhart and Ott (2006), Berninghaus, Ehrhart, Ott and Vogt (2007), Falk and Kosfeld (2003) and Goeree, Riedl and Ule (2009)).

In contrast, our experiment joins a number of recent studies that have investigated network formation with the latter type of rule, that is, when link creation requires mutual consent. Following Jackson and Watts (2002), we look at a sequential game where the network evolves one link at a time.[2] Two experimental studies – Pantz (2006) and Kirchsteiger, Mantovani, Mauleon and Vannetelbosch (2011) – examine a similar question. Our experiment differs methodologically from these two studies in two important respects. First, we consider networks of six rather than four subjects. This dramatically enlarges the individual choice set and the number of possible outcomes.[3] Second, in those experiments a pairwise-stable (PWS) network

[1] There is a somewhat larger experimental literature on equilibrium selection in network games (see e.g. Fatas, Meléndez-Jiménez and Solaz, 2010; Charness, Feri, Meléndez-Jiménez and Sutter, 2012).

[2] Alternatively, Goyal and Joshi (2006) examine a linking game with mutual consent where at every turn, each agent simultaneously proposes to *all* other agents. See, e.g., Di Cagno and Sciubba (2008) and Burger and Buskens (2009) for experimental studies of this type of game.

[3] For instance and as discussed below, with 4 players there are 64 possible networks

always existed and was the empty network, or the full network, or both. Instead, we consider four treatments with unique, multiple, or no PWS network. In the two treatments with a unique PWS network, it is neither the empty nor the full network.[4]

The results of the experiment can be summarized as follows. The analysis of *outcomes* suggests that the network formation process tends to converge to the PWS network when it exists and not to converge to any network when a PWS network does not exist. However, convergence is by no means guaranteed, especially when the PWS network involves asymmetric payoffs among subjects. When multiple PWS networks exist, subjects often manage to reach the high-payoff one.

To understand better the choices of our subjects, we then study each *single decision*. The percentage of myopically rational behavior is high. However, descriptive and regression analysis indicate four instances where this percentage is decreased. First, at the beginning of the match. This is intuitive. Subjects play at least 12 rounds, so they know it is likely that a decision taken at the beginning can be reversed later on if desired. Second, when deviations are necessary to escape the low-payoff PWS network. This reinforces the result emphasized previously that subjects look for the high-payoff network. Third, when the marginal payoff losses are small. This would be consistent with a theory of 'imperfect choice' where deviations from myopic rationality are inversely proportional to the cost of doing so. And fourth, when the myopic rational decision is to remove links. Again, this is intuitive because such an action does not require the consent of other subjects and therefore may be taken unilaterally later on if the opportunity presents.

A cluster analysis performed at the *subject level* reveals heterogeneous behavior across individuals. More than half of our subjects exhibit a sustained increase in myopic rationality as the end of the match approaches, and low myopic rationality when required to reach the high-payoff PWS network. About a quarter of the subjects have a rather erratic behavior: they deviate substantially from myopic rationality without differentiating between treatments or turns. The rest exhibit an interesting pattern, with initially low myopic rationality (consistent with costless experimentation) and a dramatic increase in myopic rationality (around 20%) when choices

and 6 minimally-connected architectures whereas with 6 players there are 32,768 possible networks and 20 minimally-connected architectures.

[4]Other differences include a benefit function from network membership that depends exclusively on the size of the network, a long sequence of choices with an average of 102 individual decisions per match, and a reshuffling of partners after each match.

are most likely to be irreversible.

Finally, we briefly compare our results with the literature. First, despite our game being more complex than Pantz (2006) and Kirchsteiger et al. (2011), we still find support for forward-looking behavior in the treatment with multiple PWS networks, just like they do. Second, the treatment with no PWS network allows us to establish that convergence is greatly facilitated by the existence of a PWS network, as predicted by Jackson and Watts (2002). Third, we show that convergence crucially depends on the architecture of the (unique) PWS network. In particular, it is more difficult to reach PWS in the treatment where different members of the network have different payoffs (e.g., the extremes vs. the center in components with 3 subjects) than in the treatment where all subjects have identical payoffs (e.g., components with 2 subjects). Fourth, our setting permits a detailed study of the determinants of myopic rational choices. We show that deviations from myopic rationality are rare but, at the same time, more prevalent when one would expect them to occur: in early turns, when marginal losses are small, and for choices that are easier to reverse later on.

The paper is organized as follows. In Section 2, we present the conceptual framework and network models that are pertinent to our experiment. Section 3 elaborates the experimental design and introduces our treatments. Then, in the following three sections, we present our analysis at the final network level (Section 4), single decision level (Section 5) and subject level (Section 6). Section 7 concludes.

2 Network environment and basic definitions

A network is a collection of links connecting "nodes" which, in our case, represent agents. A link between two agents can only form if *both* decide that it is worth forming. Each link is costly for both agents and this cost is non-transferable. Meanwhile, the benefit depends on and is a strictly increasing function of the size of the network component that an agent belongs to. We distinguish between "network" and "component". A network describes the link configurations that include the full graph (i.e., all six agents) while a component is a sub-graph in which there exists a path linking any two agents. We assume that all agents in a component receive the same benefit. Payoffs are computed as the difference between benefits and costs.

Given that link formation needs mutual consent while link destruction does not, what network is likely to emerge? A candidate is the pairwise-stable network. A network is *pairwise-stable* (PWS) if: (i) all existing links

are weakly preferred by both agents in the link and are strictly preferred by at least one of them; and (ii) all non-existing links are such that at least one of the agents on the non-existing link strictly prefers its absence (Jackson and Wolinsky, 1996). One other plausible outcome arises when agents maximize the total value of the network, that is, the sum of payoffs received by all agents. This is referred to the *efficient network* (Jackson and Wolinsky, 1996).

A network consists of minimally connected components, also called a *minimally connected* network, if the removal of any existing link increases the number of components (Bala and Goyal, 2000). In our setting, stable networks and efficient networks always consist of minimally connected components. To see why, notice that when benefits are strictly a function of the component size as in our game, in any non-minimally connected network removing a link that does not split a component will increase the net payoff of two individuals without affecting the net payoffs of any other subject.

Pairwise stability, however, does not provide a prediction of which network outcome will be reached. Jackson and Watts (2002) model a dynamic process of link formation and destruction by assuming that pairs of agents meet at random and decide whether to propose a link to each other when none exists, or sever an existing one. We call a *myopic rational* choice one in which agents make their decisions based solely on the marginal payoff resulting from the potential link they are considering (and not on the option value of forming or severing links in the future). Under myopic rationality, a link is created if both agents are weakly better-off with the link and at least one is strictly better-off. A link is severed if at least one agent is strictly better-off without the link. If all agents follow a myopic rational behavior, then the network evolves following an *improving path*.

Starting from any network, Jackson and Watts (2002, Lemma 1) show that improving paths lead to either a PWS network or a cycle. A set of networks forms a *cycle* if there is an improving path from any network to any other network in the set. A set of networks is in a *closed cycle* if no network in the set is on the improving path of a network that is not in the set. Since improving paths assume myopic rational choices, they predict for example that players will be stuck in an empty network when developing a network is initially costly but eventually beneficial. Also, there might be multiple PWS networks, some more attractive than others. In that case, depending on the starting network configuration, the improving path will lead to one PWS network or another, independently of their payoff properties.

To deal with predictions under multiplicity, we consider the possibility of farsighted behavior. We call PS-PWS a *Pareto-Superior pairwise stable*

network in which the payoffs of all agents are weakly higher (strictly for at least one of them) than in any other PWS network. If agents follow myopic rational choices, they will be stuck in any PWS network they reach, but if agents realize the long run benefits of myopically suboptimal choices, they may move away from one PWS network into a PS-PWS network.[5]

Finally but crucial for our experiment, notice that different agents in a component may obtain different payoffs (they all have the same benefits but some may have more direct links than others). This poses another multiplicity problem: even within a PWS network, agents have incentives to make or break links in order to change their final position in the network (typically trying to keep the same component size but have other agents bear most of the link costs).

3 Experimental setting and procedures

3.1 The basic configuration

Our experiment examines how well efficient networks and PWS networks predict the outcome of a dynamic linking game. We are interested in environments with a *large number of network configurations* where links are costly and mutual pairwise consent is needed to form a link but not to break it. To this end, we implement a stochastic dynamic linking game that slightly modifies the procedure proposed by Jackson and Watts (2002). We consider networks with $n = 6$ players. This means $n(n-1)/2 = 15$ possible bilateral undirected links between different players, and therefore $2^{n(n-1)/2} = 32,768$ possible networks. Notice that many networks differ from each other only on the identity of subjects in the different positions (e.g., which players are at the center of the component and which ones are at the extremes). We say that two networks have the same *network architecture* if they are identical up to a permutation of the identity of subjects. With 6 players there are 20 minimally-connected network architectures.[6]

We specifically look at a setting where all agents in a component receive the same benefits. One example of such a setting is the formation of risk-

[5] This definition is different from the notion of *pairwise farsightedly stable* network developed by Herings, Mauleon and Vannetelbosch (2009) which explicitly considers the notion of farsighted improving paths.

[6] Although six players may not seem like a large network, it is the biggest size we could manage. For comparison, Pantz (2006) and Kirchsteiger et al. (2011) have four players, which means 6 bilateral undirected links, 64 possible networks and just 6 minimally-connected network architectures.

sharing insurance networks, where individuals commit to share monetary holdings equally with linked partners (Bramoullé and Kranton, 2007). With repeated interactions between linked partners, they formally show that such an arrangement is tantamount to equal sharing within network components. Another example would be the case of club goods (such as those provided by religious or social groups, see e.g., Berman (2000)) without centralized coordination. In this case, all members benefit from the participation from an additional member, but to ensure participation may require individuals to maintain costly links with other members. We therefore deviate from much of the network experimental literature that implements the connections model where the benefits of indirect links decay with distance (Jackson and Wolinsky, 1996). One advantage of our approach is that payoffs from decisions are straightforward to calculate. This is made even simpler by the fact that we maintain the unit cost of a link to be constant both within and across treatments. By doing so, we dramatically reduce the likelihood that some decisions are made out of payoff miscalculations.

Each match consists of multiple turns and starts with an empty network. At each turn, the computer randomly assigns the six players into three pairs. Once paired, players choose their actions with respect to their partner in the pair. A new turn begins after all players have taken their actions. If all players are satisfied with the network outcome, they can collectively end the game. If we were to mimic the dynamic linking procedure of Jackson and Watts (2002), we should require matches to end only when all players have agreed on the outcome. However, this would make sessions unmanageably long. Furthermore, some agents could sabotage the experiment and choose actions randomly knowing that they always have the option to change it in the future. It would also favor those who stubbornly refuse to give up until a specific position in a specific component is reached. We therefore decided to implement a match-ending rule that provides enough opportunities for players to converge but, at the same time, allows decisions to be meaningful and the experiment to be time manageable: Subjects play for 12 turns unless all players are satisfied with the network; afterwards, each turn is the last one with probability $p = 0.2$, providing an additional $1/p = 5$ turns on average. With this probabilistic match-end rule, we hope to mitigate the last-round effects. At the same time, it allows for an interesting comparison of behavior before and after the 12th turn. Finally, notice that each turn is composed of six decisions, one for each player, providing a fairly large number of individual decisions per match ($17 \times 6 = 102$ on average, unless subjects decide to stop before).

Figure 1 shows the user interface. At each turn, players make decisions

by clicking on one of the action buttons. If a player is not linked to his partner, he chooses whether to "Propose" a link or "Pass Turn". If he is linked, he chooses whether to "Remove" a link or "Pass Turn". Once a pair of partners have taken their actions, the result is immediately displayed on the screen. Hence, when each player makes his decision, he observes the latest state of the network. Showing the latest network configuration *within a turn* allows us to cleanly determine whether each individual decision reflects a myopic rational behavior or otherwise. On the other hand, it may encourage a war of attrition, where players wait to see what others do in a turn before choosing their own action.[7]

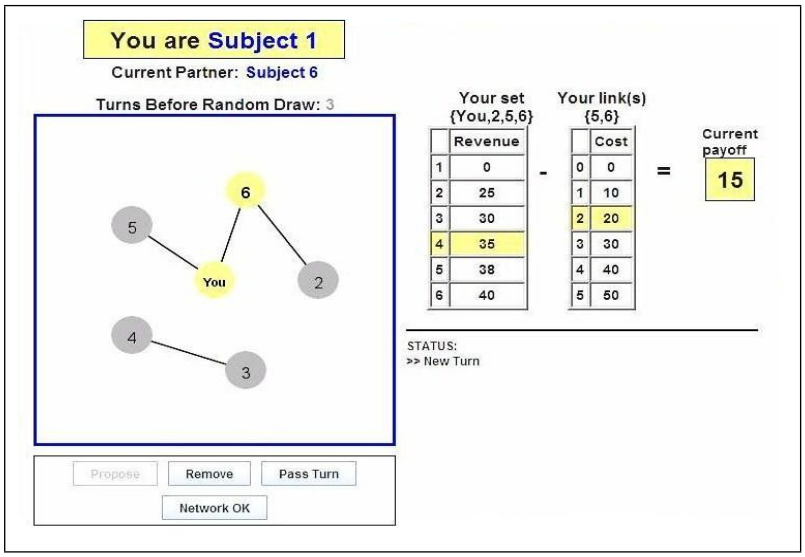


Figure 1: User interface for the linking game.

If a player is not only satisfied with the relationship with his partner but also with the overall network, instead of choosing "Pass Turn", he can choose "Network OK".[8] As mentioned above, the match immediately ends if all players within a turn choose "Network OK".

The user interface displays all the pertinent information: the subject's role, the role of the person he is currently matched with, whether the current turn is a potential terminal turn and, naturally, the current network configuration. It also displays the benefit of the subject as a function of the

[7]There is no evidence in our data of such war of attrition type of behavior.

[8]Once a player chooses "Network OK", he does not need to choose further actions until the network changes. To avoid mistakes, all of his action buttons become inactive. These buttons are immediately reactivated following a change in the network.

size of the component he is in, the cost as a function of his number of direct links, and his net payoff given the current network configuration. This succinct but comprehensive visual display allows the subject to compute rather easily not only the net value of adding or removing an existing link (i.e., the improving path) but also his payoff in any other network configuration. Finally, notice that the node representing the subject is always located at the center and labeled "You". The nodes representing the other players in a match are labeled by their roles and surround the subject's node in clockwise order at an equal distance from it. By always putting the subject's node at the center, even though the underlying connections between subjects in a match are identical, each subject sees a different graphical representation. We therefore avoid leading participants towards focal networks such as the star or wheel network.

3.2 Treatments

Since we are primarily interested in comparing observed and predicted outcomes under different scenarios, the benefits are not based on any particular functional form. With this freedom (we only impose a benefit strictly increasing in component size and a constant unit link cost), we can set the payoffs such that we have treatments with unique PWS network architectures that are different from the empty or full network, with no PWS network, and with multiple PWS networks. We also facilitate computations by maintaining the same unit link cost across treatments and only vary the benefits.

Our experiment includes four treatments. They are graphically depicted as Treatments 1, 2, 3 and 4 and are inserted at the end of the paper for reference. These figures illustrate how we determine the different PWS networks. First, we draw a "supernetwork", comprising the 20 possible minimally-connected six-node network architectures (labeled $\{A\}$ to $\{T\}$). Then we add all the arcs connecting pairs of networks that differ from one another by a single link. We put all network architectures with identical number of links in the same row and order them from top row (network with no links) to bottom row (networks with 5 links). Each network is therefore connected by an arc to one or more networks in the row above and the row below it.[9] The direction of the arc represents the improving path: forming a new link (arrow pointing to the row below) or removing an existing link (arrow pointing to the row above).

[9]Networks that are not minimally connected are necessarily off the improving paths. They are omitted unless a match ends in one of them.

The four treatments we consider differ on existence and number of PWS networks. Treatment 1 has no PWS network. There is a unique closed cycle comprised of networks $\{B, C, D, F, G, H, N\}$ within which the dynamic process is expected to stay (shaded area). Treatment 2 has four PWS networks, $\{A, I, J, K\}$, and network $\{K\}$ (in light shade) is Pareto-Superior to the other three. If players exhibit myopic rational behavior and follow the improving path, they will stay on the empty network $\{A\}$. If for some reason (a perturbation, a mistake, an experimentation, etc.) they reach any of the other PWS networks, $\{I, J, K\}$, they will also stay there. However, with some form of forward-looking behavior, players might be able to reach the Pareto-Superior PWS network $\{K\}$. Treatments 3 and 4 have unique PWS networks (shaded), $\{H\}$ and $\{L\}$, with three components of two subjects each and two components of three subjects each respectively.

Meanwhile, the efficient networks are identical across treatments: all the minimally-connected full networks, that is, $\{O, P, Q, R, S, T\}$. We choose to have the same efficient networks in all treatments to facilitate comparisons. Also, we choose several efficient networks and one of them focal ($\{T\}$, the line that comprises all agents) in order to give a fair chance to the efficient outcome. The information is summarized in Table 1.

Table 1: Efficient and PWS networks

Treatment	Benefit in a component of size						Cost per link	Efficient network(s)	PWS network(s)
	1	2	3	4	5	6			
1	0	20	30	39	42	43	15	$[6]_{\{O,P,Q,R,S,T\}}$	None
2	0	10	17	22	38	44	15	$[6]_{\{O,P,Q,R,S,T\}}$	$[1\text{-}1\text{-}1\text{-}1\text{-}1\text{-}1]_{\{A\}}$ $[5\text{-}1]_{\{I,J,K^*\}}$
3	0	29	36	41	43	44	15	$[6]_{\{O,P,Q,R,S,T\}}$	$[2\text{-}2\text{-}2]_{\{H\}}$
4	0	19	36	42	44	45	15	$[6]_{\{O,P,Q,R,S,T\}}$	$[3\text{-}3]_{\{L\}}$

The number in brackets refers to the size of each component.

* *is the Pareto-Superior PWS network (PS-PWS).*

3.3 Implementation details

The experiment was conducted in the California Social Science Experimental Laboratory at UCLA in August 2010. All experimental subjects were UCLA students. We conducted 8 experimental sessions with 12 subjects in each

session. Subjects played two sets of four treatments for a total of 8 matches in each session. We shuffled the order of the treatments such that: (i) the orders of the treatments in the first half and the second half of each session are different; (ii) no two sessions have identical treatment sequences; and (iii) each treatment is implemented in exactly two sessions for each order in the sequence. This shuffling was done to neutralize the possible effects from the ordering of the treatments within a session. In total, we obtained 128 match observations (32 matches for each treatment) from 96 subjects (12 subjects in 8 sessions).

With 12 subjects, there are always 2 groups in each session. To introduce anonymity in game play, after each match we reshuffled subjects into new groups and assigned a new role (1 to 6) to each subject. Each session lasted for between 90 and 120 minutes. No subject took part in more than one session. Participants interacted exclusively through computer terminals without knowing the identities of the subjects they played with. Before the paid matches, instructions were read aloud and two practice matches were played to familiarize participants with the computer interface and procedure. After that, participants had to complete a quiz to ensure they understood the rules of the experiment.

At the end of each match, subjects obtained a payoff based on the size of the component they were in (benefit) and the number of direct links (cost). Participants were endowed with experimental tokens and they could earn or lose tokens. At the end of the session, the payoffs in tokens accumulated from all experimental games were converted into cash, at the exchange rate of 4 tokens = \$1. Participants received a show-up fee of \$5, plus the amount they accumulated during the paid matches. Payments were made in cash and in private. Matches lasted for between 13 and 36 turns, with an average of 16.8 turns. There was a significant spread in winnings: including the show-up fee, participants earned between \$11 and \$43 with an average of \$29. A copy of the instructions is included in Appendix B.

4 Network outcomes

We first analyze the general properties of the network formation game, most notably convergence and stability of the final configuration.

4.1 Network convergence: efficiency vs. stability

Result 1 *Subjects rarely end up in components that are not minimally connected.*

We begin by noting that players show understanding of the basic tenets of the experimental game. Table 2 summarizes the network outcomes of the four treatments. We first focus on whether players end up in networks with components that are not minimally connected. Remember that removing a link that does not reduce the component size strictly increases the payoffs of both agents involved without affecting any other. The second column in Table 2 suggests that players understand this principle. Only 5 matches out of 128 (or 4%) end up in a network that is not minimally connected. The result is all the more remarkable that non-minimally connected networks are sometimes necessary intermediate steps towards farsighted goals.

Table 2: Summary of network outcomes

Treatment	Not Min. Conn.	Efficient	PWS	PS-PWS	Closed cycle
1	2 (6.3%)	3 (9.4%)	-	-	21 (65.6%)
2	1 (3.1%)	5 (15.6%)	3* (9.4%)	14† (43.8%)	-
3	0 (0.0%)	0 (0.0%)	15 (46.9%)	-	-
4	2 (6.3%)	2 (6.3%)	10 (31.3%)	-	-

*$N = 32$ for each treatment; * networks $\{A, I, J\}$; † network $\{K\}$.*

Result 2 *The process tends to converge to the PWS network when a PWS exists and not to converge to any network when no PWS network exists. It rarely leads to an efficient network.*

Before discussing the result, we need an operational definition of convergence. Callander and Plott (2005) argue that a network has converged if it maintains the same configuration in the last T turns before the end of the game. Alternatively, we can use the "Network OK" action. In this case, a network is said to have converged if all six players chose "Network OK" or if five players did so and the sixth was either unable or unwilling to unilaterally change the network based on her previous actions.

The first definition may "over-detect" convergence since it includes cases where the lack of change is merely the result of how pairs were randomly

assigned in the last T turns. The second definition may "under-detect" convergence. Indeed, a player may delay choosing "Network OK" hoping to benefit from a possible mistake or deviation by another player.

Table 3 presents the rate of convergence under the first definition with $T = 3$ and under the second definition.[10] Under the first definition, the convergence rate is lowest in Treatment 1 (around 20%), where there is no PWS network. For the treatments with unique or multiple PWS networks, convergence is above 50%. Convergence under the second definition is always low (16% or less) indicating that this notion is probably too strict.

Table 3: Network convergence

Treatment	No change in last 3 turns†	Network OK
1	7 (21.9%)	1 (3.1%)
2	20 (62.5%)	3 (9.4%)
3	17 (53.1%)	5 (15.6%)
4	18 (56.3%)	3 (9.4%)

$N = 32$ for each treatment; †*Includes "Network OK".*

We now go back to Table 2 and look at how well pairwise stability predicts the final network. The graphical illustration of Treatments 1 to 4 elaborates on this information: it displays the number of final outcomes in each network architecture $\{A\}$ to $\{T\}$ both conditional on no change in the last 3 turns (labeled C) and unconditional on convergence (labeled U). In the absence of a PWS network (Treatment 1), 66% of the matches end in a network within the closed cycle.[11] More interestingly and ignoring convergence for now, 53%, 47% and 31% of the final networks in Treatments 2, 3 and 4 are PWS. Finally, the dynamic formation process rarely leads to

[10] $T = 3$ is arbitrary. It corresponds to 18 individual decisions which seems reasonably large. With larger T, convergence decreases but the qualitative conclusions remain largely the same ($T = 5$ is not presented for brevity but it is available from the authors).

[11] Matches stay within the closed cycle most of the time: in 56% of all the moves after Turn 6 subjects remain in a network within the closed cycle. The result should not be overemphasized as the closed cycle is rather large (7 out of 20 network architectures).

any of the efficient networks.[12]

We next examine the following question: When stable networks exist, how well do they predict the outcome conditional on convergence? Hereafter, we employ the operational definition of convergence as the lack of change in the last $T = 3$ turns. Table 4 suggests a mixed picture. For Treatments 2 and 3, more than half of the convergent networks are PWS networks. For Treatment 4, only 22% are (Treatment 1 is not included in this analysis as it predicts no convergence). The difference between Treatments 3 and 4 is intriguing and deserves further study. A possible reason is the asymmetry of players' payoffs *within* components. Indeed, with [2-2-2] all subjects earn equal amounts and have little room for improvement. By contrast, with [3-3] the players in the center of the component may deviate, hoping to reach the same network but with someone else bearing the two link costs.

Table 4: Summary of network outcomes conditional on convergence

Treatment	Efficient	PWS	PS-PWS	N
2	4 (20.0%)	1* (5.0%)	11 (55.0%)	20
3	0 (0.0%)	9 (52.9%)	-	17
4	1 (5.6%)	4 (22.2%)	-	18

* *Includes networks* $\{A, I, J\}$.

In Appendix A.1 we provide a complementary analysis based on the *distance* between the observed and predicted (efficient, closed cycle, PWS and PS-PWS) outcomes, both conditional and unconditional on convergence. The conclusions are similar. The distance is smaller between the observed outcome and the networks in the closed cycle (Treatment 1), the PS-PWS network (Treatment 2) and the PWS network (Treatment 3) than between the observed outcome and any of the efficient networks in Treatments 1, 2 and 3. Again, the result is less clearcut for Treatment 4.

[12]The treatment that ends with the efficient network the most is Treatment 2 (16%). However, in this treatment, all efficient networks that became the final outcome are just one link away from the PS-PWS.

4.2 Network stability: myopic vs. farsighted

Result 3 *Agents are able to "coordinate" away from the myopic improving path to reach the higher return, PS-PWS network.*

As elaborated in the previous section, the strict use of myopic improving paths may lead to networks with low returns for all subjects, especially in the presence of scale economies. Treatment 2 implements such scale economies: agents with one link obtain positive payoffs for components of size 3 and above and agents with two links obtain positive payoffs for components of size 5 and above. Thus, starting from the empty network $\{A\}$, the improving path will stay there. Meanwhile, $\{K\}$, the PS-PWS network has a [5-1] architecture.

Tables 2 and 4 show that the dynamic link formation process did lead by and large to the high return network. Indeed, 44% (unconditional on convergence) and 55% (conditional on convergence) of matches in Treatment 2 matches ended in the PS-PWS network. By contrast, only 9% (unconditional) and 5% (conditional) ended in one of the other three PWS networks.

The results are in line with Pantz (2006) who found that many of her games did reach the forward-looking network architecture. Our setup, however, is more complex for players. First, we have 20 minimally-connected architectures. Starting from the empty PWS network $\{A\}$, players must go through *three* non-improving paths before they reach networks $\{F, G\}$, and only from there they may converge to the PS-PWS network $\{K\}$.[13] And second, the sequence of link decisions in Pantz (2006) is fixed and known *ex ante* by all players, so subjects can compute the subgame-perfect equilibrium of the game. Instead, our subjects face random pairing at each round.

4.3 Payoffs

Table 5 presents the average sum of payoffs obtained by subjects in each treatment, also called "network value". These values are compared with the average value of the networks in the closed cycle (for Treatment 1), with the values of the PWS and PS-PWS networks (for Treatments 2, 3 and 4), and with the values of the efficient networks (for all treatments). We find substantial losses compared to the payoffs that could be collectively obtained: the empirical payoffs are between 46% and 71% of the payoffs generated by any of the efficient networks. In fact, the observed payoffs are

[13] As can be seen from the graphical illustration of Treatment 2, all the arcs representing the improving paths in rows 2, 3 and 4 are directed upwards (towards $\{A\}$ and away from $\{K\}$), hence the need for at least three non-myopic rational choices.

smaller but close to the payoffs in the PS-PWS network for Treatment 2 and to the payoffs in the unique PWS network for Treatments 3 and 4. For Treatment 1, the payoff is 50% higher than the average payoff in the closed cycle. The results are similar when we consider only the empirical payoffs of the convergent networks.

Table 5: Summary of network values

Treatment	Obtained All	Obtained Convergent	PWS	PS-PWS	Closed cycle	Efficient
1	63.13 (26.58)	-	-	-	43.1 †	108
2	52.91 (44.68)	62.70 (41.53)	0 ‡	70	-	114
3	81.47 (12.33)	79.35 (12.80)	84	-	-	114
4	85.38 (17.88)	84.78 (16.51)	96	-	-	120

Standard deviations in parenthesis

† *Unweighted average payoffs of all networks in the cycle.*

‡ *We only consider the empty network* $\{A\}$.

4.4 Summary

Subjects in our game understand the strategic nature of network formation and avoid non-minimally connected components. Network efficiency does not appear to be an important motivation in any treatment. The existence of one or several PWS networks is necessary for convergence: when no PWS network exists (Treatment 1) the process hardly converges. At the same time, it is not sufficient: convergence is weaker in Treatment 4 than in Treatments 2 or 3. Finally, subjects are reasonably forward-looking when this is needed to reach the high-paying PS-PWS network.

5 Individual decisions

Having analyzed aggregate outcomes, we now study each individual decision. At each turn, each subject in a pair must choose to either "act" or "pass". If subjects in the pair are initially unlinked, acting implies proposing a link and passing implies remaining unlinked. If, on the contrary, subjects are initially

linked, acting implies removing a link and passing implies remaining linked. We are interested in the extent to which decisions are myopic rational in each of these four cases and for each treatment.

5.1 Descriptive statistics

Table 6 summarizes the proportion of myopic rational actions across turns. We organize the data into four groups of turns. We use the last certain turn that players get unless everyone agrees on the network outcome (Turn 12) as a natural point to partition and further split each of these partitions into two.

Table 6: Myopic rationality of individual decisions

	Turns†				Mean difference t-test‡	
	[1-6]	[7-12]	[13-18]	≥ 19	[7-12] v. [1-6]	[13-18] v. [7-12]
A. All	0.741	0.819	0.894	0.910	0.000	0.000
	(0.438)	(0.385)	(0.308)	(0.286)		
B. By decision problem						
i. Stay unlinked (Pass)	0.374	0.392	0.571	0.647	0.292	0.000
	(0.484)	(0.489)	(0.496)	(0.485)		
ii. Stay linked (Pass)	0.882	0.878	0.877	0.846	0.613	0.505
	(0.323)	(0.328)	(0.328)	(0.362)		
iii. Remove link (Act)	0.663	0.837	0.916	0.940	0.000	0.000
	(0.473)	(0.369)	(0.277)	(0.238)		
iv. Propose link (Act)	0.956	0.967	0.979	0.967	0.144	0.109
	(0.206)	(0.179)	(0.144)	(0.180)		
C. By treatment						
Treatment 1	0.829	0.826	0.885	0.944	0.566	0.000
	(0.377)	(0.379)	(0.320)	(0.229)		
Treatment 2	0.563	0.856	0.905	0.862	0.000	0.001
	(0.496)	(0.351)	(0.294)	(0.345)		
Treatment 3	0.764	0.788	0.895	0.919	0.082	0.000
	(0.425)	(0.409)	(0.306)	(0.274)		
Treatment 4	0.811	0.805	0.890	0.969	0.644	0.000
	(0.392)	(0.397)	(0.313)	(0.175)		

†Standard deviations in parenthesis.

‡P-values of one-sided t-test of whether mean in latter turns is greater than in earlier turns.

This split captures behaviors at different stages. First, players attempt to get familiar with the current match and try different strategies which, with high probability, can be reversed later if desired (Turns [1-6]). Then, players adjust their behavior as the last certain turn approaches (Turns [7-12]). After that, players enter the random stopping phase where, presumably, they behave under the assumption that matches can be terminated at any time (Turns [13-18]). Finally, Turns 19 and above are set in a another category because the sample size is dramatically reduced as turns advance and the sample becomes non-representative of the population.[14]

Although formal tests are presented in the regression analysis of section 5.2, Table 6 is instructive. Panel A suggests that decisions are more myopic rational as players get closer to the end of the match, with jumps statistically significant at the 0.1% level between [1-6] and [7-12] and also between [7-12] and [13-18].

Panel B investigates myopic rationality by type of decision. We examine choices under four mutually exclusive conditions, namely when the rational action is: (i) to pass and remain unlinked; (ii) to pass and remain linked; (iii) to remove an existing link; and (iv) to propose a new link. By comparing conditions (i) with (ii), and (iii) with (iv), we find evidence that individuals deviate more from improving paths in decisions that reduce the number of links than in decisions that increase the number of links. By comparing conditions (i) with (iii), and (ii) with (iv), individuals appear to deviate more from improving paths by failing to act when they should than by acting when they should not.[15] However, our regressions below suggest that this last result does not hold once we control for individual fixed effects and marginals payoff from myopic rational choices.

Panel C displays myopic rationality across treatments. In all four treatments, players are significantly less myopic rational before Turn 12 than after it at the 0.1% level. Interestingly, the difference in myopic rationality between [1-6] and [7-12] described in Panel A is entirely driven by Treatment 2. This supports the hypothesis of forward-looking behavior since in Treatment 2 and only in that treatment, subjects need to play against myopic rationality at least three times in order to escape the basin of attraction of the zero payoff PWS network $\{A\}$. A graphical illustration of the results in panels B and C is presented in Appendix A.2.

[14] The split is arbitrary. Similar results are obtained if the first and third partition are changed marginally. The key issue is to introduce a separation around turn 12.

[15] A set of t-tests (not reported here) confirms that for each turn group, the mean differences in myopic rationality both between conditions (i) and (iii) and between conditions (ii) and (iv) are negative and statistically significant at the 0.1% level.

Finally, Table 7 presents the number of instances in which individuals choose to "enter", "stay" and "leave" a PWS network, broken down by treatment and turn. The last column reports the total number of turns in that set. We observe that individuals are prone to leave the PWS network in Turns [1-12], but this tendency is dramatically reduced when each turn can be final, especially for Treatment 3. Surprisingly, even though Treatment 2 has a predictable outcome (44% of matches end up in the PS-PWS network, see Table 2), individuals move away from the PS-PWS network $\{K\}$ after Turn 12 more frequently than they do in the other treatments, possibly due to the existence of other PWS networks as well. Also, subjects leave the zero payoff PWS network $\{A\}$ in turn 1 and never come back to it.

Table 7: Movements to and from PWS networks

Turns	Enter	Stay	Leave	Total turns
Turns [1-12]				
Treatment 2 - $\{A\}$ †	0	–	–	384
Treatment 2 - $\{K\}$	28	51	19	384
Treatment 3	30	69	24	384
Treatment 4	7	17	4	384
Turns [$\geq$ 13]				
Treatment 2 - $\{A\}$	0	–	–	177
Treatment 2 - $\{K\}$	14	67	9	177
Treatment 3	9	44	0	143
Treatment 4	8	19	1	128

† Excludes the initial choice

5.2 Regression analysis

To study more thoroughly individual decisions, we regress the probability that an individual chooses the myopic rational action on the attributes of the problem. That is, for each treatment we estimate a linear probability model (LPM) for the following specification:

$$\mathbf{P}(Y_{nt}^{ij} = 1 \mid \mathbf{X_{nt}^{ij}}, c_n) = \beta_0 + \mathbf{X_{nt}^{ij}}\boldsymbol{\beta} + c_n \tag{1}$$

where Y_{nt}^{ij} indicates whether the action is myopic rational, $\mathbf{X_{nt}^{ij}}$ captures the attributes that move individual n from network i to j in the supernetwork at turn t. Meanwhile, c_n captures the unobservable characteristics of the

individual n which may affect how she makes decisions. We do not assume that the unobservable individual characteristics are independent from the attributes of the decisions, and hence, implement an individual fixed effects specification. We also cluster standard errors by session.

We choose the fixed-effects linear probability model (LPM) instead of a logit model because it is easier to interpret the marginal effects for the former, especially with regards to the interaction terms (Ai and Norton, 2003).[16] At the end of the section, we briefly discuss some extensions and alternative representations.

We can use the regression framework to investigate the four types of decisions described in Panel B of Table 6. Consider the following specification:

$$\begin{aligned} \mathrm{E}(Y_{nt}^{ij} \mid \mathbf{X}) = \beta_0 + \beta_1 \cdot morelink_{ij} + \beta_2 \cdot act_{ij} \\ + \beta_3 \cdot (morelink_{ij} \times act_{ij}) + \varepsilon \end{aligned} \tag{2}$$

where $morelink_{ij}$ and act_{ij} are dummy variables and ε is the residual. The variable $morelink_{ij}$ takes on a value of 1 if between networks i and j the network with more links gives the individual a higher payoff; the variable act_{ij} takes on a value of 1 if the myopic rational choice is to act.

Under the LPM, the interpretation of these β-coefficients is straightforward. The coefficient β_0 captures the probability that an individual does not propose a link in accordance to the myopic rational strategy (M. rat.). Similarly, $\beta_0 + \beta_1$ captures the probability that an individual does not remove a link in accordance to the myopic rational strategy. Table 8 provides interpretations for the different combinations of coefficients.

Table 8: The regression coefficients and the types of decision problems

	Interpretation	$more\ link_{ij}$	act_{ij}	Function
i.	$\mathbf{P}$(M. rat. = 1 | M. rat. = Stay unlinked)	0	0	β_0
ii.	$\mathbf{P}$(M. rat. = 1 | M. rat. = Stay linked)	1	0	$\beta_0 + \beta_1$
iii.	$\mathbf{P}$(M. rat. = 1 | M. rat. = Remove)	0	1	$\beta_0 + \beta_2$
iv.	$\mathbf{P}$(M. rat. = 1 | M. rat. = Propose)	1	1	$\beta_0 + \beta_1 + \beta_2 + \beta_3$

This specification allows us to explore how the nature of the problem affects game play. We also include three sets of additional variables (and the individual fixed effects) to explore individual strategies. The extended

[16] We do not consider the fixed-effects probit model given its known bias (Greene, 2004).

models are, therefore, variations based on the following specification:

$$\begin{aligned} \mathrm{E}(Y_{nt}^{ij} \mid \mathbf{X}) = \beta_0 &+ \beta_1 \cdot morelink_{ij} + \beta_2 \cdot act_{ij} + \beta_3 \cdot (morelink_{ij} \times act_{ij}) \\ &+ \gamma \cdot mpay_{ij} + \delta_q \cdot chdist(q)_{ij} \\ &+ \textstyle\sum_{t=1}^{4} \chi_t \cdot turn_sp(t) + c_n + \varepsilon \end{aligned} \tag{3}$$

First, we investigate whether the size of marginal payoffs affects the deviations from the improving path. The variable $mpay_{ij}$ contains the marginal payoff from making a myopic rational choice between networks i and j. With a myopic strategy, the sign of the decision (a loss vs. a gain) should matter but not the magnitude of the loss or the gain. However, if we assume imperfect choices (like in the Quantal Response Equilibrium model of McKelvey and Palfrey (1995) for example) it is reasonable to think that deviations are less likely to occur when marginal losses are large.

Second, we explore the possibility that individuals follow the shortest distance towards the efficient line network (Eff.Line) or the Pareto-Superior PWS network (PS). The variable $chdist(q)_{ij}$ denotes the change in the geodesic distance to network $q \in \{\text{Eff.Line, PS}\}$ if an individual takes the myopic rational decision in choosing between networks i and j. Each of these variables can take a value of 1, 0, or -1 and they are included one at a time in the regression. A negative coefficient on $chdist(q)$ indicates that all else the same, players are more likely to choose a myopic rational action if it moves them closer to network q.

Finally, we control for possible turn effects using a linear spline on the turn variables, $turn_sp$, with knots at turns 6, 12, and 18.[17] The knot choices mimic the turn grouping we did for the descriptive analysis.

Result 4 *Individual decisions are more myopic rational after Turn 13 than before Turn 12.*

We first perform a test for pooling for all of our specifications to investigate whether there is a structural change after Turn 12. In all but one specification, we can reject the null hypothesis that the coefficients before and after Turn 12 are equal at 1% significance; in all cases, we can reject the hypothesis at 5% significance (see Table 15 in Appendix A.3 for details).

We therefore analyze the two sets of turns separately. We begin by examining the extent to which improving paths drive individual behaviors. If improving paths were the sole driver of network evolution, the constant

[17] Hence, the variable $turn_sp(1)$ is the spline for Turns [1-6], $turn_sp(2)$ is for Turns [7-12], $turn_sp(3)$ is for Turns [13-18] and $turn_sp(4)$ is for Turns 19 and greater.

terms in all specifications would be one and the coefficients on all other variables would be zero. Table 9 presents the regression results of our basic model with individual fixed effects. The constant terms are high but significantly lower than one, and the coefficients of the other variables are significantly different from zero, suggesting deviations from the improving paths.

To better study deviations, we included estimates of the linear combinations of the coefficients for the constant term, *morelink*, *act*, and *morelink*$\times$ *act*. These linear combinations are derived from Table 8 to allow immediate comparisons of the probabilities that individuals make myopic rational choices for the different decision problems.

Pairwise comparisons of estimates confirm that, all else the same, individuals are more myopic rational in Turns [$\geq$ 13] than in Turns [1-12]. Of the 16 combinations of treatments and decision problems, the point estimates are always larger in later turns with only one exception: Treatment 2 when the myopic rational choice is to propose.

Result 5 *In early turns, individuals deviate from improving paths by maintaining excessive links (over-proposing and not removing redundant links). In later turns, individuals deviate mainly by not removing redundant links.*

Panel B of Table 6 suggests that subjects keep, if anything, too many links. We hypothesized that the asymmetry of the linking game may explain this behavior. Since link formation requires mutual agreement while removal does not, one possible strategy would be to form and maintain some redundant links early on. As the game approaches the end, individuals begin to unilaterally remove some of them.

We find some evidence supporting this hypothesis in our regressions. As shown in Table 9, in Turns [1-12] the coefficient for myopic rationality in all four treatments is highest when the myopic rational action is to stay linked (ii), followed by propose a link (iv), stay unlinked (i), and remove a link (iii). For Turns [$\geq$ 13], the least myopic rational decision is still by far to remove a link (iii), except for Treatment 2. The order of the other coefficients are somewhat perturbed although it is difficult to make strong conclusions since most coefficients are very high (90% and above).

Result 6 *The size of marginal payoffs affects the likelihood of a deviation from myopic rationality in early turns for Treatments 1, 3 and 4 and in all turns for Treatment 2.*

Table 9: FE LPM on myopic rationality: effect of type of decision

	Turns [1-12]				Turns [≥ 13]			
	Tr. 1 (1)	Tr. 2 (2)	Tr. 3 (3)	Tr. 4 (4)	Tr. 1 (5)	Tr. 2 (6)	Tr. 3 (7)	Tr. 4 (8)
morelink [β_1]	0.168***	0.256***	0.233***	0.144***	0.040	0.015	0.097*	0.072*
	(6.51)	(25.39)	(22.96)	(6.51)	(1.28)	(1.25)	(2.96)	(2.43)
act [β_2]	-0.265**	-0.267*	-0.277***	-0.512***	-0.374**	-0.074	-0.168**	-0.418***
	(-4.28)	(-2.64)	(-5.96)	(-18.75)	(-3.56)	(-1.35)	(-3.65)	(-5.45)
morelink × act [β_3]	0.173**	0.182	0.263**	0.444***	0.287	-0.068	0.201**	0.335**
	(3.81)	(1.86)	(4.79)	(17.47)	(2.21)	(-0.95)	(4.02)	(5.38)
Constant [β_0]	0.809***	0.664***	0.731***	0.816***	0.942***	0.916***	0.878***	0.921***
	(46.84)	(69.11)	(127.02)	(65.97)	(46.10)	(57.06)	(89.13)	(51.57)
Individual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Linear combinations:								
i. β_0	0.809***	0.664***	0.731***	0.816***	0.942***	0.916***	0.878***	0.921***
	(46.84)	(69.11)	(127.0)	(65.97)	(46.10)	(57.06)	(89.13)	(51.57)
ii. $\beta_0+\beta_1$	0.977***	0.920***	0.964***	0.960***	0.981***	0.931***	0.976***	0.993***
	(48.56)	(64.59)	(114.4)	(77.36)	(70.81)	(56.75)	(32.78)	(36.17)
iii. $\beta_0+\beta_2$	0.544***	0.397***	0.454***	0.303***	0.568***	0.842***	0.711***	0.503***
	(11.57)	(4.240)	(10.26)	(16.29)	(6.020)	(18.73)	(16.16)	(6.990)
iv. $\beta_0+\beta_1+\beta_2+\beta_3$	0.885***	0.835***	0.950***	0.892***	0.894***	0.789***	1.009***	0.909***
	(57.83)	(35.82)	(49.59)	(60.15)	(30.18)	(14.40)	(25.63)	(19.83)
N	2304	2304	2304	2304	972	1062	858	768
Adj. R^2	0.142	0.125	0.256	0.267	0.264	0.171	0.242	0.228

t statistics in parentheses. Standard errors are clustered at the session level.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 10: FE LPM on myopic rationality: effect of marginal payoff

	Turns [1-12]				Turns [≥ 13]			
	Tr. 1 (1)	Tr. 2 (2)	Tr. 3 (3)	Tr. 4 (4)	Tr. 1 (5)	Tr. 2 (6)	Tr. 3 (7)	Tr. 4 (8)
morelink	0.311***	0.428**	0.303**	0.455*	0.120	0.053	0.124	0.127*
	(5.66)	(4.32)	(5.15)	(3.20)	(1.38)	(1.14)	(1.39)	(2.40)
act	-0.309*	0.249*	-0.381***	-0.613**	-0.063	0.227	-0.390*	0.084
	(-2.98)	(3.39)	(-6.18)	(-4.02)	(-0.37)	(2.19)	(-2.71)	(0.27)
morelink × act	0.213	-0.251**	0.454***	0.579**	-0.043	-0.531*	0.524*	-0.241
	(1.90)	(-3.83)	(9.54)	(3.83)	(-0.24)	(-3.01)	(2.78)	(-0.69)
mpay [γ_0]	0.011*	0.034**	0.015*	0.026*	0.009	0.011**	0.010	0.006
	(2.89)	(5.24)	(3.35)	(2.74)	(2.24)	(4.65)	(2.09)	(2.19)
mpay × morelink [γ_1]	-0.012*	-0.024*	-0.014*	-0.024*	-0.008	-0.002	-0.009	-0.005
	(-3.20)	(-3.37)	(-3.30)	(-2.45)	(-1.59)	(-0.64)	(-1.73)	(-1.31)
mpay × act [γ_2]	0.008	-0.048**	0.011	0.017	-0.033	-0.027	0.027	-0.043
	(0.95)	(-4.86)	(1.75)	(1.57)	(-1.33)	(-2.28)	(1.90)	(-1.50)
mpay × morelink × act [γ_3]	-0.008	0.045**	-0.013	-0.018	0.035	0.038*	-0.032	0.048
	(-0.82)	(4.07)	(-2.23)	(-1.47)	(1.46)	(2.64)	(-2.11)	(1.55)
Constant	0.659***	0.012	0.534***	0.408*	0.767***	0.736***	0.793***	0.798***
	(10.74)	(0.29)	(9.33)	(2.94)	(11.60)	(18.08)	(16.89)	(16.57)
turn_sp(1) & turn_sp(2)	Yes	Yes	Yes	Yes	No	No	No	No
turn_sp(3) & turn_sp(4)	No	No	No	No	Yes	Yes	Yes	Yes
chdist(EffLine)	Yes	No	Yes	Yes	Yes	No	Yes	Yes
chdist(PS)	No	Yes	No	No	No	Yes	No	No
Individual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Linear combinations:								
i. γ_0	0.011*	0.034**	0.015*	0.026*	0.009	0.011**	0.010	0.006
	(2.89)	(5.24)	(3.35)	(2.74)	(2.24)	(4.65)	(2.09)	(2.19)
ii. $\gamma_0+\gamma_1$	-0.001	0.010**	0.001	0.003*	0.001	0.009**	0.001	0.001
	(-0.94)	(5.34)	(1.18)	(2.87)	(0.52)	(4.32)	(0.43)	(0.36)
iii. $\gamma_0+\gamma_2$	0.019*	-0.014	0.026**	0.044***	-0.024	-0.016	0.037*	-0.037
	(3.16)	(-1.39)	(4.17)	(5.86)	(-0.91)	(-1.23)	(3.03)	(-1.31)
iv. $\gamma_0+\gamma_1+\gamma_2+\gamma_3$	-0.000	0.007**	-0.002	0.003*	0.003	0.020**	-0.004	0.005
	(-0.03)	(5.29)	(-0.82)	(2.79)	(1.21)	(5.36)	(-0.64)	(1.14)
N	2304	2304	2304	2304	972	1062	858	768
Adj. R^2	0.154	0.375	0.300	0.294	0.291	0.282	0.270	0.246

t statistics in parentheses. Standard errors are clustered at the session level.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

With a myopic rational strategy, the size of the marginal payoffs should be irrelevant: individuals would choose actions that give them non-zero gain, irrespective of their size. To investigate whether this is the case, we implement the extended specification of (3). We then interact the payoff variable with the interactions between *morelink* and *act* to capture differential effects of marginal payoffs across different decision problems.

The results of the regressions are presented in Table 10. We linearly combine the coefficients for the payoff variables to explore the heterogeneity of the payoff-size effects across decision problems. We use a strategy similar to the way we linearly combined in Table 8 the coefficients of the *morelink*, *act* and *morelink* $\times$ *act* to examine the myopic rationality of the different decision problems. Hence, for example, γ_0 measures how the size of the marginal payoff affects the probability that individuals myopic-rationally stay unlinked; $\gamma_0 + \gamma_1$ measures how the payoff size influences the probability that individuals myopic-rationally stay linked, and so on.

For Turns [1-12] in Treatments 1, 3 and 4, myopic rational choices are positively correlated with the size of marginal payoffs if the myopic rational choice is to reduce the number of links (cases (i) and (iii)). Most of these coefficients lose their significance in Turns [$\geq$ 13]. Meanwhile, for Treatment 2, all of the payoff coefficients are significant in both [1-12] and [$\geq$ 13] except when the myopic rational choice is to remove a link.

These results provide additional insights on how individuals deviate from improving paths. For Treatments 1, 3 and 4, the evidence suggests that individuals take the opportunity loss from removing a link more seriously than that from staying unlinked. For Treatment 2, subjects take all losses into consideration. As explored next, they presumably see those (potential though not necessarily realized) losses as an intermediary step towards the PS-PWS network.

Result 7 *Individual strategies are strongly suggestive of forward-looking behavior.*

Finally, we examine the crucial question of forward-looking behavior in Treatment 2. The analysis at the network outcome level suggests that individuals tend to reach the PS-PWS network (Result 3). We examine whether this conclusion is also supported at the individual-level. Table 11 presents the extended regression based on (3). Two pieces of evidence corroborate the notion that individuals are forward looking.

First, remember that individuals in Treatment 2 would need to violate myopic rationality in the first few turns to escape the zero-payoff PWS

Table 11: FE LPM on the likelihood of myopic rational action

	Turns [1-12]					Turns [≥ 13]				
	Tr. 1	Tr. 2		Tr. 3	Tr. 4	Tr. 1	Tr. 2		Tr. 3	Tr. 4
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
morelink [β_1]	0.156***	0.121***	0.092***	0.129***	0.120**	0.018	0.027	0.016	0.016	0.061
	(6.22)	(7.94)	(5.41)	(6.56)	(4.82)	(0.80)	(1.90)	(1.16)	(0.25)	(2.12)
act [β_2]	-0.242**	-0.254	-0.206	-0.258***	-0.469***	-0.349*	0.007	0.018	-0.149**	-0.410**
	(-4.32)	(-2.18)	(-2.00)	(-6.77)	(-15.60)	(-2.99)	(0.12)	(0.37)	(-3.54)	(-5.12)
morelink × act [β_3]	0.161**	0.232	0.186	0.293**	0.438***	0.265	-0.160	-0.174	0.188*	0.319**
	(4.39)	(1.93)	(1.68)	(4.96)	(15.49)	(1.92)	(-1.96)	(-2.19)	(3.01)	(5.32)
turn_sp(1) †	-0.006	0.079***	0.079***	0.012	0.003					
	(-1.01)	(10.96)	(10.86)	(2.03)	(0.30)					
turn_sp(2) †	0.012*	-0.000	0.000	0.009	0.005					
	(2.73)	(-0.10)	(0.08)	(2.00)	(1.85)					
turn_sp(3) †						0.017*	0.002	0.003	0.006	0.016
						(2.81)	(0.64)	(0.73)	(0.91)	(1.82)
turn_sp(4) †						0.003	-0.007	-0.008	-0.007	0.001
						(1.23)	(-2.29)	(-2.35)	(-1.80)	(0.09)
chdist(Eff. Line)	-0.025*		-0.086**	-0.056**	-0.037*	-0.018		-0.036*	-0.034	0.006
	(-3.42)		(-5.37)	(-5.23)	(-2.71)	(-1.18)		(-2.60)	(-1.93)	(0.60)
chdist(PS)		-0.071***					-0.038*			
		(-5.81)					(-2.98)			
Constant [β_0]	0.773***	0.157***	0.197***	0.621***	0.707***	0.843***	0.693***	0.713***	0.857***	0.837***
	(19.37)	(7.01)	(7.17)	(15.74)	(14.06)	(28.07)	(24.16)	(32.38)	(17.34)	(18.81)
mpay	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Linear combinations:										
β_0	0.773***	0.157***	0.197***	0.621***	0.707***	0.843***	0.693***	0.713***	0.857***	0.837***
	(19.37)	(7.009)	(7.167)	(15.74)	(14.06)	(28.07)	(24.16)	(32.38)	(17.34)	(18.81)
$\beta_0+\beta_1$	0.929***	0.278***	0.289***	0.750***	0.827***	0.861***	0.720***	0.730***	0.872***	0.898***
	(23.24)	(9.824)	(9.472)	(21.81)	(22.47)	(53.26)	(29.77)	(32.58)	(8.398)	(15.55)
$\beta_0+\beta_2$	0.531***	-0.097***	-0.009***	0.363***	0.238***	0.495***	0.700***	0.732***	0.707***	0.426***
	(16.28)	(-0.831)	(-0.0975)	(5.535)	(5.785)	(5.184)	(14.97)	(17.31)	(11.76)	(5.469)
$\beta_0+\beta_1+\beta_2+\beta_3$	0.848***	0.257***	0.268***	0.785***	0.796***	0.778***	0.566***	0.574***	0.911***	0.807***
	(45.19)	(6.557)	(6.374)	(30.34)	(37.74)	(24.07)	(7.394)	(7.862)	(7.650)	(11.60)
N	2304	2304	2304	2304	2304	972	1062	1062	858	768
Adj. R^2	0.186	0.381	0.388	0.319	0.311	0.353	0.341	0.342	0.344	0.336

t statistics in parentheses

† Spline coefficients are for the slope of the intervals. Standard errors are clustered at the session level.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

network $\{A\}$. However, once these initial "barriers" have been overcome, individuals can reach the PS-PWS network $\{K\}$ by following the improving paths. We therefore expect an increase in myopic rationality within the first six turns. As shown in Table 11, the spline for the first six turns, $turn_sp1$, is positive and significant for Treatment 2 and only for that treatment. All other turn spline variables for Treatment 2 are not significant.

Second, the coefficient for the distance change to the PS-PWS network, *chdist(PS)*, is negative and significant in both early and late turns (columns (2) and (7) in the table). It suggests that individuals may be implementing strategies that move them closer to the PS-PWS network. Notice, however, that the impact of this variable is similar to that of the distance to the efficient-line network variable, *chdist(Eff.Line)* (columns (3) and (8) in the table). Because the two networks differ by a mere one link, we cannot identify which network is the individuals' ultimate goal. However the evidence regarding the convergent network outcomes for Treatment 2 (in Table 4) suggests that the PS-PWS architecture is the ultimate goal.

5.3 Model predictions

We next examine the capacity of our empirical model to *predict* actions. For this exercise, we include the regressors of the basic model and the marginal payoff variable interacted with *morelink*, in line with our previous findings. All the variables are interacted with $turn > 12$ to account for coefficient differences before and after Turn 12. We also include only the spline for the first six turns, and interact it with a Treatment 2 indicator variable to account for players' forward-looking behavior in that treatment. For the prediction model, we do not include the individual fixed effects.

The results of the Logit regressions used for out-of-sample predictions are relegated to Appendix A.4. Figure 2 graphically depicts the plot of the actual (dashed line) and out-of-sample predicted (solid line) proportion of myopic rational choices for each treatment in the first 18 turns. The model predicts aggregate behavior well. As expected, for Treatment 2 it overestimates myopic rationality at early stages, when deviations are necessary to reach the PS-PWS network, but it predicts actual behavior accurately after Turn 6. The model slightly under-predicts myopic rationality in early turns of Treatment 3 and slightly over-predicts myopic rationality in late turns of Treatment 4.

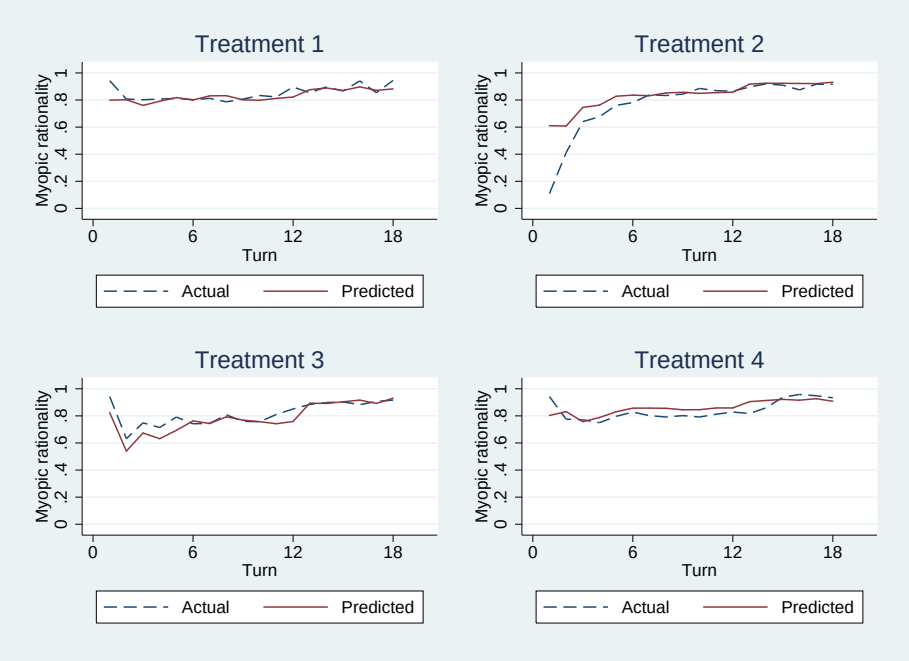


Figure 2: Out-of-sample logit prediction by treatment

5.4 Extensions and alternative representations

We have focused on the LPM in the regression. As a robustness check, we also estimated fixed-effects logit models. The results for the analogue of Tables 9 and 11 in these alternative estimations are presented in Appendix A.5. For the most part, the estimates are qualitatively similar, except for the coefficients of the interactions between *morelink* and *act*.

We also entertain the possibility that experience matters. Remember that each treatment is played twice in a session. We examine differences in behavior between the first and second time a treatment is encountered by running a test of pooling. The results are presented in Appendix A.6. In general, we find little evidence of differences in behavior. The most notable difference is Treatment 2 where, in their second encounter, subjects are less myopic rational when it prescribes to remove or propose a link.

5.5 Summary

The individual-choice analysis suggests that subjects play less myopically rational in turns with sure future than in random ending turns and also less

myopically rational when deviations are necessary to escape the low-payoff PWS network. Deviations tend to take the form of excessive links, possibly because they can be removed unilaterally, although proving this hypothesis would require further work. Finally, deviations are also more prevalent the smaller the marginal payoff losses, as expected in a behavioral theory where mistakes depend inversely on loss magnitudes. Overall and with some interesting exceptions, the analysis provides support for convergence to PWS networks through myopic rational choices, except when multiple PWS networks exist in which case subjects exhibit the forward-looking attitude necessary to reach high payoffs.

6 Heterogeneity across subjects

So far we have studied choices at the network outcome and single decision levels. One question that remains unanswered is the degree of heterogeneity between subjects. A simple way to address this question is to determine how often *each subject* plays the myopic rational strategy.

Figure 3 plots the cumulative distribution function (CDF) of myopic rational behavior of each subject by treatment. A steep CDF reflects homogeneity across subjects whereas a right shift captures a more myopic rational behavior on aggregate. In Treatments 1, 3 and 4, behavior is myopic rational and homogeneous: 80% of subjects play the myopic rational strategy 75% of the time or more. In Treatment 2, behavior is slightly less myopically rational and more heterogeneous. A Kolmogorov-Smirnov test confirms this observation: the CDF of Treatment 2 is different from the CDF of Treatments 1, 3 and 4 at the 1% level, whereas no statistical difference is observed between the CDFs in Treatments 1, 3 and 4 at the 10% level.

Heterogeneity can also be studied by searching for clusters of people (as in Camerer and Ho (1999) for example). This allows us to quantify the degree of similarity of subjects' behavior within and between clusters. Given the documented difference in behavior between early and late turns and also between Treatment 2 and Treatments 1, 3 and 4, we use these four variables to cluster the subjects. There are many clustering methods but they usually require the number of clusters and the clustering criterion to be set ex-ante rather than endogenously optimized. Mixture models, on the other hand, treat each cluster as a component probability distribution. Thus, the choice of the model and the number of clusters is made using Bayesian statistical methods (Fraley and Raftery, 2002). We implement model-based clustering analysis with the Mclust package in R (Fraley and

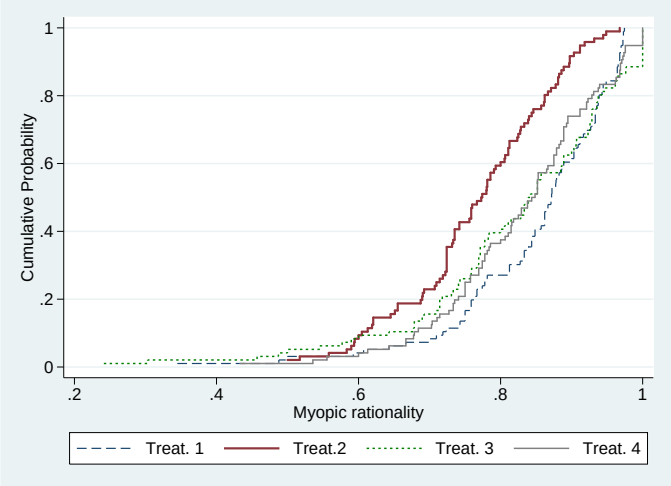


Figure 3: Empirical CDF of myopic rationality by treatment

Raftery, 2009). A maximum of nine clusters are considered for up to ten different models and the combination that yields the maximum Bayesian Information Criterion (BIC) is chosen.

For our multidimensional data, the model that maximizes the BIC provides *five* clusters. Table 12 shows the frequencies of subjects in each cluster, listed from low to high according to the general frequency of their myopic rational behavior. It also displays their average earnings.

Result 8 *There is substantial heterogeneity across individuals. Some subjects consistently follow improving paths except when deviations are needed to reach the PS-PWS network whereas others exhibit a rather erratic behavior.*

Clusters are clearly differentiated in: (i) total level of myopic rationality, (ii) difference in behavior between Treatment 2 and the other treatments, and (iii) difference in behavior between early and late turns. Subjects in cluster 1 are lowest on all three dimensions, followed by subjects in cluster 2. Their erratic behavior and undifferentiated choice across treatments and turns indicates that one-quarter of our subjects have difficulties in understanding some of the strategic aspects of the game. Subjects in cluster 3 start with low myopic rationality but have the sharpest increase after turn 12 (around 20%). The near perfectly myopic rational behavior in later stages of Treatments 1-3-4 suggests that, contrary to subjects in clusters 1 and 2, these individuals understand the incentives in our game and use early turns to look for an advantageous position. Clusters 4 and 5, which

Table 12: Clustering based on myopic rational behavior

Cluster	Treatments 1-3-4		Treatment 2		Earnings	N
	[1-12]	[≥ 13]	[1-12]	[≥ 13]		
1	0.678	0.735	0.617	0.645	86.5	10
	(0.132)	(0.178)	(0.134)	(0.226)		
2	0.786	0.850	0.708	0.814	95.8	13
	(0.060)	(0.060)	(0.099)	(0.082)		
3	0.779	0.955	0.645	0.844	83.7	19
	(0.125)	(0.026)	(0.147)	(0.099)		
4	0.843	0.909	0.746	1.000	96.6	46
	(0.076)	(0.076)	(0.113)	(0.000)		
5	0.870	0.975	0.771	0.881	113.8	8
	(0.038)	(0.034)	(0.059)	(0.054)		
Total	0.808	0.897	0.710	0.897		96
	(0.104)	(0.104)	(0.127)	(0.145)		

Standard deviations in parenthesis

comprise more than half of the population, are a portrait of rational individual maximization in a strategic game: very high myopic rationality in later turns of all treatments, a somewhat lower myopic rationality in early turns of Treatments 1-3-4 (possibly due to costless experimentation), and a significantly lower myopic rationality in early turns of Treatment 2 (necessary to escape the low-payoff PWS network). The main difference between the two clusters lies in the late turn treatments where myopic rationality is highest: Treatment 2 for cluster 4 and Treatments 1-3-4 for cluster 5.

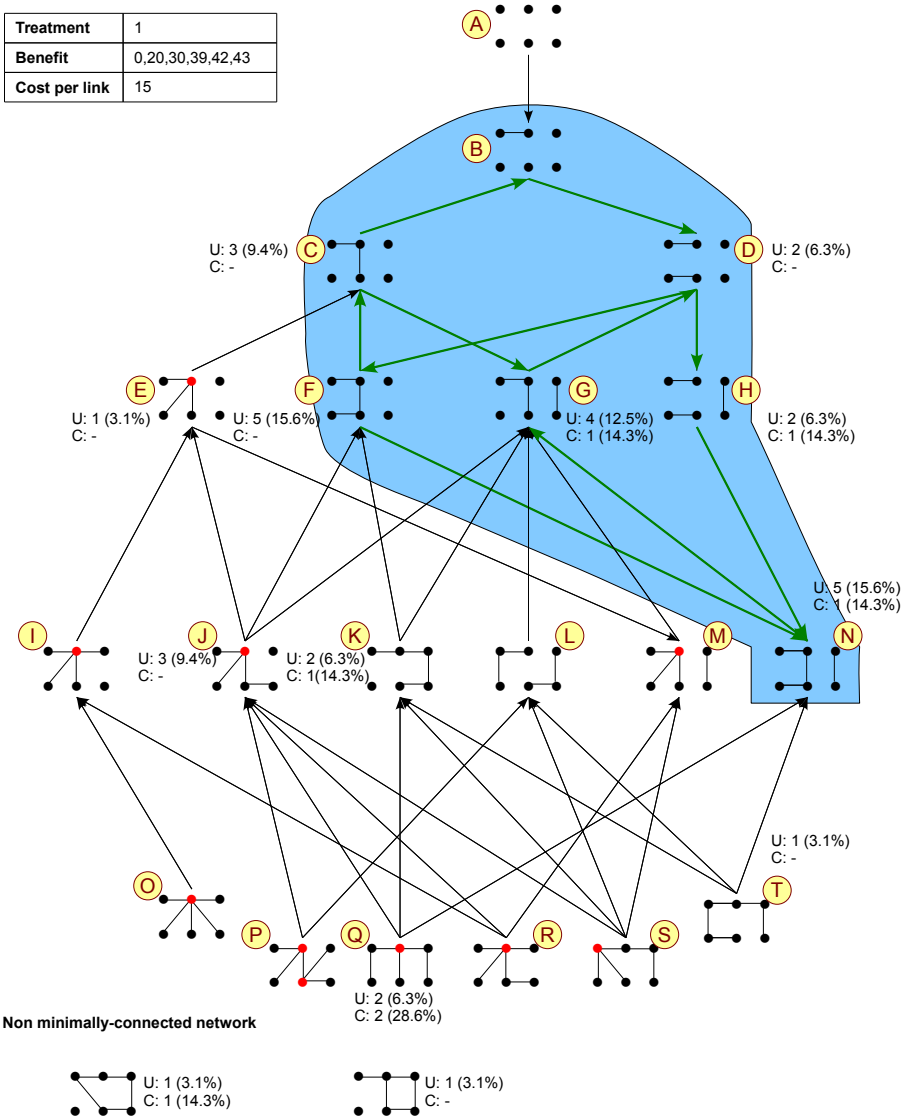
Earnings are correlated with behavior, with clusters 1 and 5 near the bottom and top of the distribution. However, the mapping between payoffs and myopic rationality is not monotonic. This can be expected for several reasons. First, because PWS networks do not necessarily generate the highest payoffs. Second, because a subject's payoff depends on the behavior of the five other players in the network and their final position in the component. Rational subjects may end up bearing a larger number of links.

7 Conclusion

The paper has studied the dynamic formation of social networks. We found that subjects rarely consider the total value of the network as a key criterion when making their decisions. Instead, choices are roughly consistent with individual maximization of payoffs. Many subjects exhibit forward-looking behavior if (and only if) this strategy leads to a PS-PWS configuration. Interestingly, myopic rationality is less prevalent at the margin when actions are reversible (early turns), when marginal payoff losses are smaller and when actions involve excessive links that can be removed unilaterally later on. There is, however, a significant heterogeneity in behavior across subjects.

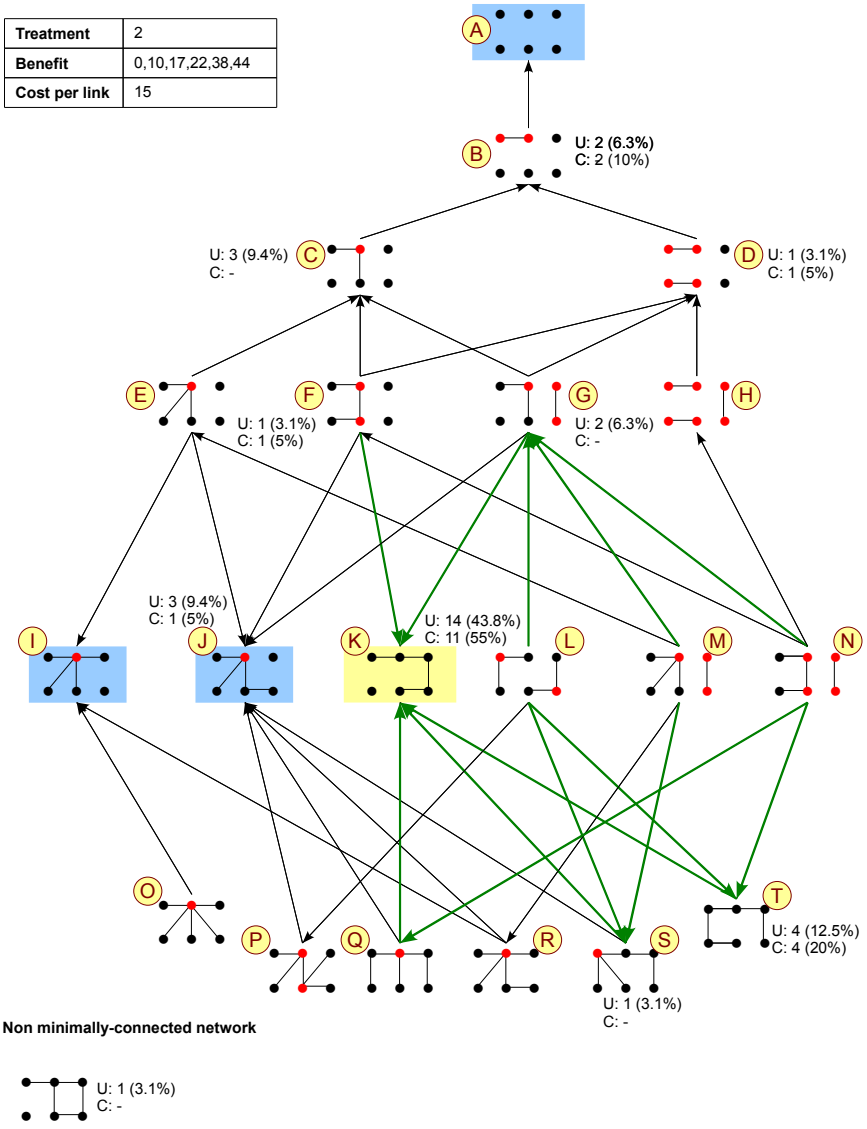
Despite the recent advances, there is still much to learn about network formation, both theoretically and experimentally. On the theory front, it would be desirable to incorporate behavioral imperfections into existing models. The tendency observed in our data towards fewer deviations from myopic rationality as marginal losses increase and as matches get closer to the end strongly suggests that players optimize subject to imperfect choice, imperfect foresight and/or imperfect information processing capacity. To our knowledge, however, no model has yet been developed to capture these frictions. On the experimental front, ecological validity is a concern. Indeed, we feel that our cost and benefit representation of adding and removing links captures the essence of social networks in an excessively abstract way. The use of laboratory studies in the field or laboratory studies that exploit social technologies (facebook, twitter, second life, etc.) would add a more realistic dimension to the network formation problem without compromising the controlled environment of the laboratory.

Treatment 1



Notes. Letters refer to all minimally-connected network architectures. Numbers next to each network refer to the frequency (and percentage) that the process ends in that network: U = Unconditional on convergence, C = Conditional on no change in the last 3 turns. Networks that are part of the closed cycle are inside the shaded region.

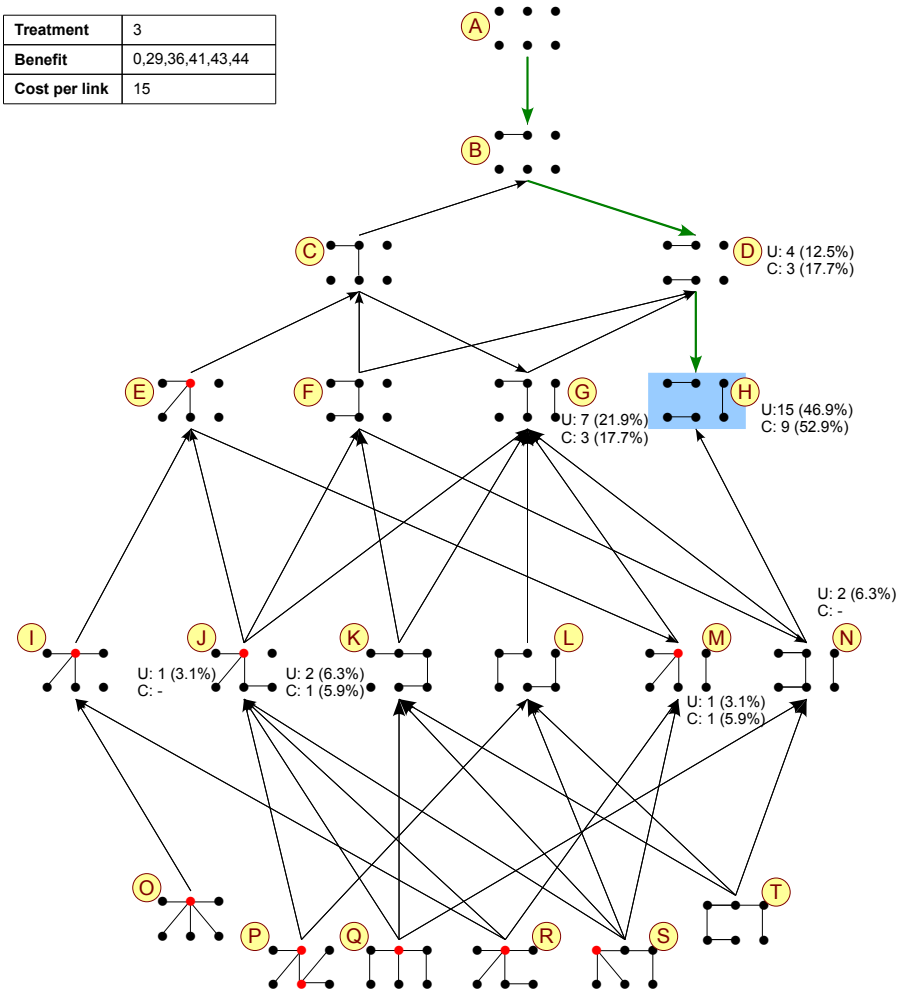
Treatment 2



Notes. *Letters refer to all minimally-connected network architectures. Numbers next to each network refer to the frequency (and percentage) that the process ends in that network: U = Unconditional on convergence, C = Conditional on no change in the last 3 turns. PWS networks are shaded (light shade for the PS-PWS network).*

Treatment 3

Treatment	3
Benefit	0,29,36,41,43,44
Cost per link	15

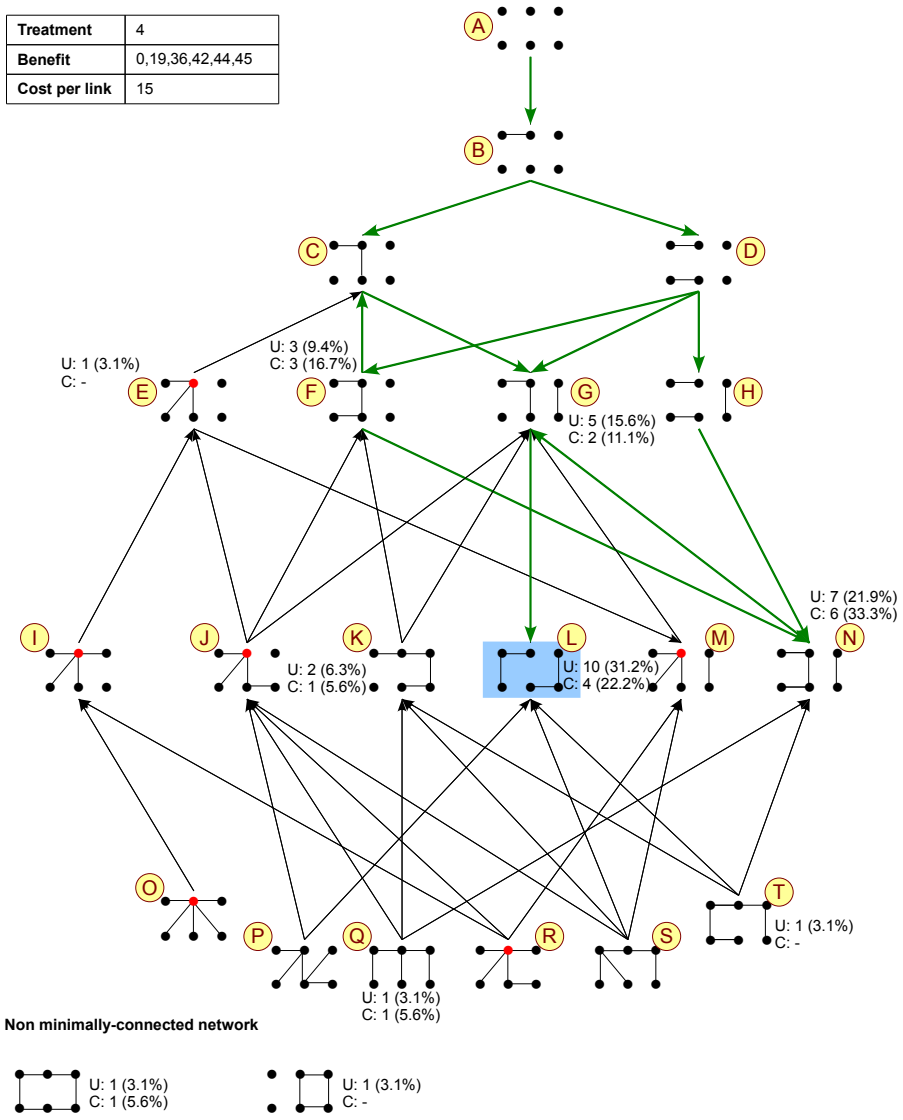


Non minimally-connected network

Notes. Letters refer to all minimally-connected network architectures. Numbers next to each network refer to the frequency (and percentage) that the process ends in that network: U = Unconditional on convergence, C = Conditional on no change in the last 3 turns. PWS network is shaded.

Treatment 4

Treatment	4
Benefit	0,19,36,42,44,45
Cost per link	15



Notes. Letters refer to all minimally-connected network architectures. Numbers next to each network refer to the frequency (and percentage) that the process ends in that network: U = Unconditional on convergence, C = Conditional on no change in the last 3 turns. PWS network is shaded.

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Appendices

A Additional analyses (for online publication only)

A.1 Geodesic distance to efficient and PWS networks

In this section, we provide a complementary study of the difference between observed and predicted outcomes to the one presented in section 4.1. More precisely, we calculate the shortest (or "geodesic") distance between the resulting networks and the closest network in the closed cycle (for Treatment 1) or the PWS networks (for Treatments 2, 3 and 4) as well as the distance between the resulting networks and the efficient networks. For the latter, we separately calculate the distance to the closest of all the efficient networks $\{O, P, Q, R, S, T\}$ and to the line network $\{T\}$.[18] Table 13 shows that for Treatments 1 and 3, the distance to the closed cycle and the PWS network respectively is substantially shorter than the distance to the efficient networks. For Treatment 2, the distance is shorter to the PS-PWS network but longer to the PWS empty network $\{A\}$. For Treatment 4, however, the distance from the PWS network is equal to the distance from the efficient line network and longer than the distance from the closest efficient network, suggesting a larger dispersion in behavior.

Table 13: Average distances from outcomes

Treatment	PWS	PS-PWS	Closed cycle	Shortest efficient	Efficient line
1	-	-	0.41 †	1.66	2.03
2	3.66 ‡	0.91	-	1.41	1.66
3	1.00	-	-	1.94	2.06
4	1.41	-	-	1.28	1.41

† *We calculate the distance to the closest network in the cycle.*
‡ *We only consider the empty network* $\{A\}$.

Table 14 presents the average distance between outcomes and predicted networks conditional on convergence. For Treatments 2 and 3, the results provide further support for convergence to the PS-PWS network and the

[18] If agents were to aim at the efficient network, the line network is the most likely outcome since it distributes payoffs most equally. For example $\{O\}$, which is never played in our experiment, is efficient but requires one player to form 5 links and therefore bear significant payoff losses (30 to 32 tokens depending on the treatment).

unique PWS network respectively. For Treatment 4, the distance from the convergent network to the efficient network is lower than to the PWS network. This result (as well as that in Table 13) comes from the fact that most of the network outcomes in this treatment, both conditional and unconditional on convergence, are split almost equally between the PWS network $\{L\}$ and network $\{N\}$. Since the distance between these two networks is two and both of them are at a distance of one to the efficient networks, the distance from the stable network and the efficient networks are similar. It is difficult to infer from the outcomes alone where the formation processes is leading toward. In our analysis of individual decisions (Result 5), we provide a plausible explanation for our findings here.

Table 14: Average distance from outcomes conditional on convergence

Treatment	PWS	PS-PWS	Shortest efficient	Efficient line
2	3.75 *	0.75	1.25	1.35
3	0.88	-	2.06	2.18
4	1.61	-	1.28	1.39

* *We only consider the empty network* $\{A\}$.

A.2 Myopic rationality by treatment and decision problem

Figure 4 further illustrates the results in Panels B and C of Table 6. We plot for each treatment the proportion of myopic rational behavior across turns when passing is rational (B1) and when acting is rational (B2). In all treatments, players maintain more links in all turns than would be observed if they played myopically rational all the time. The gap is bigger and the variation larger for decisions where acting is myopic rational (figures on the right), although the difference narrows as the match nears its end.

A.3 Pooled FE LPM on probability of myopic rational choices

Table 15 presents the fixed effects pooled linear probability model to investigate if there is a difference in myopic rational choices before and after Turn 12.

A.4 Logit model for out-of-sample predictions

Table 16 presents the logit model used for the out-of-sample predictions.



Figure 4: Myopic rationality by treatment and decision problem

Table 15: Pooled FE LPM on the likelihood of myopic rational action

	Treatment 1		Treatment 2			Treatment 3		Treatment 4	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1(turn>12)	0.130***	0.063	0.263***	0.486***	0.472***	0.169**	0.204*	0.105***	0.018
	(5.54)	(1.31)	(31.72)	(9.16)	(9.58)	(4.07)	(2.67)	(6.32)	(0.32)
morelink	0.163***	0.153***	0.253***	0.142***	0.109**	0.236***	0.138***	0.145***	0.123***
	(7.13)	(6.62)	(27.72)	(7.94)	(5.36)	(22.35)	(5.92)	(6.81)	(5.72)
... × **1**(turn>12)	-0.121**	-0.125***	-0.231***	-0.108**	-0.089**	-0.162**	-0.119*	-0.088**	-0.069**
	(-4.16)	(-5.49)	(-15.03)	(-4.45)	(-3.75)	(-4.23)	(-2.60)	(-4.73)	(-3.66)
act	-0.274**	-0.255**	-0.273*	-0.234	-0.182	-0.294***	-0.277***	-0.524***	-0.483***
	(-4.38)	(-4.45)	(-2.77)	(-2.12)	(-1.91)	(-6.56)	(-7.61)	(-18.38)	(-15.46)
... × **1**(turn>12)	-0.087	-0.084	0.194	0.233*	0.198	0.077	0.079	0.103	0.076
	(-1.06)	(-0.85)	(2.32)	(2.56)	(2.34)	(1.26)	(1.54)	(1.63)	(1.18)
morelink × act	0.185**	0.186**	0.188	0.202	0.152	0.265**	0.291**	0.458***	0.454***
	(3.83)	(4.65)	(1.96)	(1.77)	(1.45)	(4.90)	(5.37)	(17.77)	(16.66)
... × **1**(turn>12)	0.079	0.056	-0.291*	-0.416**	-0.384**	-0.046	-0.078	-0.130*	-0.142*
	(0.86)	(0.50)	(-3.07)	(-4.03)	(-3.57)	(-0.70)	(-1.26)	(-2.52)	(-3.12)
mpay		0.003*		0.015***	0.013***		0.005*		0.005**
		(2.67)		(9.26)	(10.84)		(2.88)		(4.94)
... × **1**(turn>12)		-0.000		-0.002	-0.001		-0.002		-0.003
		(-0.06)		(-0.52)	(-0.23)		(-0.91)		(-1.40)
turn		0.005		0.030***	0.031***		0.010*		0.004
		(1.23)		(9.00)	(10.97)		(3.20)		(1.93)
... × **1**(turn>12)		0.002		-0.034***	-0.035***		-0.007		0.007
		(0.51)		(-7.53)	(-8.83)		(-1.47)		(2.05)
chdist(Eff. Line)		-0.024**			-0.096***		-0.055**		-0.034*
		(-3.68)			(-5.96)		(-5.21)		(-2.39)
... × **1**(turn>12)		0.010			0.046*		0.030*		0.040*
		(0.61)			(2.92)		(2.65)		(3.02)
chdist(PS)				-0.087***					
				(-6.94)					
... × **1**(turn>12)				0.039*					
				(2.89)					
Constant	0.811***	0.736***	0.663***	0.286***	0.327***	0.732***	0.635***	0.818***	0.704***
	(49.21)	(21.15)	(70.98)	(11.99)	(12.77)	(59.12)	(20.63)	(57.31)	(21.23)
Individual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
P(pooling)	0.004	0.002	0.000	0.000	0.000	0.023	0.004	0.003	0.001
N	3276	3276	3366	3366	3366	3162	3162	3072	3072
Adj. R^2	0.169	0.176	0.154	0.331	0.336	0.245	0.270	0.253	0.264

t statistics in parentheses. Standard errors are clustered at the session level.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 16: Logit regression

	Treat. 1 (1)	Treat. 2 (2)	Treat. 3 (3)	Treat. 4 (4)
1(turn>12)	1.831***	1.669***	2.193***	1.487***
	(4.35)	(4.24)	(3.67)	(3.46)
morelink	2.817***	3.761***	3.246***	3.090***
	(7.95)	(9.62)	(9.13)	(8.47)
... × **1**(turn>12)	-1.961**	-1.323*	-2.179**	-1.737***
	(-2.67)	(-1.97)	(-3.16)	(-3.39)
act	-1.861***	-1.763***	-1.499***	-1.465***
	(-12.04)	(-21.76)	(-8.52)	(-7.87)
... × **1**(turn>12)	-0.139	-0.515	-0.705*	-0.433
	(-0.57)	(-1.80)	(-2.49)	(-1.34)
morelink × act	1.244***	-0.0600	0.803***	0.595*
	(4.97)	(-0.18)	(3.80)	(2.45)
... × **1**(turn>12)	-1.353**	-0.363	-0.578	-0.725*
	(-2.99)	(-1.13)	(-1.43)	(-2.19)
turn_sp(1) †	0.233***	0.0902*	0.206***	0.243***
	(8.62)	(2.11)	(6.90)	(7.43)
... × **1**(Treatment 2)	-0.0520**		-0.0479*	-0.0749***
	(-2.95)		(-2.36)	(-7.72)
mpay	0.117***	0.0825***	0.143***	0.109***
	(6.93)	(5.33)	(5.52)	(6.36)
... × **1**(turn>12)	-0.0686*	-0.0511	-0.0756	-0.0291
	(-2.14)	(-1.56)	(-1.46)	(-0.76)
mpay × morelink	-0.0863**	-0.0876***	-0.125***	-0.0944**
	(-3.29)	(-3.70)	(-3.93)	(-3.01)
... × **1**(turn>12)	0.112*	0.0823**	0.110*	0.0701
	(2.10)	(2.64)	(2.27)	(1.54)
Constant	-1.206***	-0.0529	-1.459***	-1.114***
	(-6.44)	(-0.22)	(-7.40)	(-6.06)
Observations	9600	9510	9714	9804
Pseudo R^2	0.206	0.173	0.190	0.183

t statistics in parentheses. Standard errors clustered at the session level.
† Spline coefficients are for the slope of the intervals.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

A.5 Logit estimation

In Tables 17 and 18 we present the analogue of Tables 9 and 11 using a fixed-effects logic estimation. Note that the sign and significance of the *morelink* × *act* interaction-term coefficients often differ between the LPM and logit models. This, however, does not necessarily indicate that the two models contradict each other because, unlike in linear models (such as LPM), the coefficients on the interaction terms in non-linear models (such as logit) do not easily translate into their marginal effects (for a detailed discussion, see Ai and Norton (2003)).

A.6 Effect of experience

Table 19 presents the results regarding the effect of experience on behavior. For the basic specifications, we only find evidence of a difference in behavior for Treatment 2. With additional variables, the hypothesis of no difference in behavior is rejected for Treatment 4. Looking at the individual coefficients, we find that in Treatment 2, coefficients that are significantly different in the second half of the sessions are those for *act* and *morelink* × *act*. For Treatment 4, none of the individual coefficients are significantly different in the second half of the sessions at 5% significance, and only one coefficient, namely *turn*, is significantly different at 10% significance.

Table 17: FE Logit on myopic rationality: effect of type of decision

	Turns [1-12]				Turns [$\geq$ 13]			
	Tr. 1 (1)	Tr. 2 (2)	Tr. 3 (3)	Tr. 4 (4)	Tr. 1 (5)	Tr. 2 (6)	Tr. 3 (7)	Tr. 4 (8)
morelink	2.878***	1.674***	5.044***	1.921***	2.126*	0.329	-‡	1.808*
	(5.61)	(8.54)	(5.00)	(6.15)	(2.04)	(0.93)		(2.39)
act	-1.385***	-1.199***	-1.365***	-2.608***	-2.896***	-0.964*	-1.619***	-2.854***
	(-8.46)	(-7.27)	(-7.50)	(-14.13)	(-8.05)	(-2.02)	(-3.96)	(-7.68)
morelink $\times$ act	-0.910	0.547	-1.677	1.321***	0.062	-0.651	3.481***	0.651
	(-1.67)	(1.89)	(-1.61)	(3.52)	(0.06)	(-1.06)	(4.93)	(0.76)
Constant	1.573***	0.719***	1.204***	1.682***	3.322***	2.958***	3.080***	2.804***
	(14.88)	(9.66)	(9.08)	(13.85)	(11.17)	(12.10)	(9.59)	(11.08)
Individual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ln σ_u^2								
Constant	-0.966***	-1.636***	0.114	-0.456	0.426	0.279	0.972**	-0.303
	(-3.36)	(-4.96)	(0.50)	(-1.74)	(1.16)	(0.79)	(2.64)	(-0.53)
N	2304	2304	2304	2304	972	1062	858	768

t statistics in parentheses.

† Spline coefficients are for the slope of the intervals. ‡ Dummy variable dropped due to perfect collinearity.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 18: FE Logit on the likelihood of myopic rational action

	Turns [1-12]					Turns [≥ 13]				
	Tr. 1	Tr. 2		Tr. 3	Tr. 4	Tr. 1	Tr. 2		Tr. 3	Tr. 4
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
morelink	2.869***	1.177***	0.998***	4.420***	1.939***	1.883	0.840*	0.605	-‡	1.899*
	(5.56)	(5.28)	(4.42)	(4.35)	(5.92)	(1.78)	(2.02)	(1.42)		(2.47)
act	-1.220***	-1.326***	-1.114***	-1.305***	-2.249***	-2.587***	-0.022	0.195	-1.214**	-2.699***
	(-7.00)	(-7.00)	(-5.77)	(-6.70)	(-11.41)	(-6.88)	(-0.04)	(0.36)	(-2.70)	(-6.81)
morelink × act	-1.020	1.035**	0.802*	-1.421	1.174**	-0.257	-1.974**	-2.137**	2.372**	0.271
	(-1.84)	(3.16)	(2.42)	(-1.35)	(2.91)	(-0.23)	(-2.74)	(-2.94)	(3.13)	(0.31)
mpay	0.029*	0.105***	0.094***	0.046**	0.074***	0.051*	0.171***	0.161***	0.064	0.049*
	(2.31)	(8.50)	(7.65)	(3.17)	(4.83)	(2.35)	(7.36)	(7.01)	(1.96)	(1.98)
turn spline: before 6†	-0.051	0.361***	0.354***	0.097	-0.012					
	(-0.95)	(7.74)	(7.59)	(1.83)	(-0.20)					
turn spline: b/w 6 & 12†	0.107**	0.045	0.050	0.103**	0.063					
	(2.94)	(1.20)	(1.34)	(2.71)	(1.63)					
chdist(Eff. Line)	-0.162*		-0.471***	-0.398***	-0.265***	-0.280		-0.540***	-0.449*	0.154
	(-2.38)		(-6.80)	(-4.75)	(-3.47)	(-1.89)		(-3.82)	(-2.27)	(0.95)
chdist(PS)		-0.308***					-0.468**			
		(-4.59)					(-3.09)			
turn spline: b/w 12 & 18†						0.227*	0.038	0.043	0.059	0.308**
						(2.49)	(0.45)	(0.52)	(0.62)	(3.07)
turn spline: after 18†						0.040	-0.068	-0.072	-0.079	0.220
						(0.72)	(-1.61)	(-1.70)	(-0.83)	(0.63)
Constant	1.238***	-2.038***	-1.783***	0.323	0.540	2.049***	0.798	1.013*	2.856***	1.355**
	(4.55)	(-10.59)	(-8.89)	(1.14)	(1.74)	(4.74)	(1.81)	(2.35)	(5.34)	(2.84)
Individual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ln σ_u^2										
Constant	-0.802**	-0.791**	-0.658**	0.447*	-0.269	0.473	0.923**	0.887**	1.483***	-0.399
	(-2.94)	(-3.00)	(-2.59)	(2.02)	(-1.06)	(1.31)	(2.93)	(2.78)	(3.99)	(-0.65)
N	2304	2304	2304	2304	2304	972	1062	1062	858	768

t statistics in parentheses

† Spline coefficients are for the slope of the intervals. ‡ Dummy variable dropped due to perfect collinearity.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 19: Pooled FE LPM on the likelihood of myopic rational action

	Treatment 1		Treatment 2			Treatment 3		Treatment 4	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1(second half)	-0.051	-0.073	-0.025	-0.024	-0.022	-0.020	0.009	-0.031	-0.094
	(-1.19)	(-1.54)	(-0.95)	(-0.37)	(-0.37)	(-0.88)	(0.21)	(-0.76)	(-1.24)
morelink	0.098**	0.090*	0.175***	0.121***	0.087*	0.184***	0.098*	0.098*	0.075*
	(4.65)	(3.19)	(10.07)	(6.39)	(3.29)	(7.83)	(2.88)	(3.25)	(3.09)
... × **1**(second half)	0.063	0.050	0.023	0.010	0.012	0.013	0.013	0.046	0.044
	(1.53)	(1.18)	(0.77)	(0.33)	(0.38)	(0.50)	(0.35)	(1.19)	(1.35)
act	-0.300*	-0.289*	-0.136	-0.021	0.024	-0.255***	-0.250***	-0.491***	-0.457***
	(-3.43)	(-3.25)	(-1.98)	(-0.31)	(0.45)	(-6.17)	(-6.03)	(-6.52)	(-5.89)
... × **1**(second half)	0.003	0.028	-0.222**	-0.279***	-0.261**	-0.069	-0.021	-0.032	-0.034
	(0.03)	(0.24)	(-3.71)	(-5.73)	(-4.13)	(-0.96)	(-0.24)	(-0.28)	(-0.29)
morelink × act	0.197*	0.200*	0.031	-0.111	-0.159*	0.224***	0.261***	0.404**	0.389**
	(2.80)	(2.89)	(0.46)	(-1.48)	(-2.80)	(6.05)	(8.67)	(4.91)	(5.06)
... × **1**(second half)	0.020	0.005	0.196*	0.313***	0.299**	0.085	0.044	0.065	0.104
	(0.21)	(0.04)	(3.17)	(5.45)	(4.66)	(1.25)	(0.60)	(0.51)	(0.84)
mpay		0.000		0.020***	0.018***		0.005*		0.004*
		(0.28)		(9.95)	(9.21)		(2.49)		(2.43)
... × **1**(second half)		0.003*		-0.006	-0.006		-0.004		-0.001
		(2.41)		(-1.59)	(-2.22)		(-1.08)		(-0.41)
turn		0.007***		0.010**	0.010**		0.012**		0.003
		(8.07)		(3.56)	(3.53)		(4.02)		(1.39)
... × **1**(second half)		-0.002		0.005	0.005		0.000		0.007
		(-1.25)		(0.87)	(0.97)		(0.12)		(2.32)
chdist(Eff. Line)		-0.015			-0.104**		-0.033		-0.027
		(-1.00)			(-4.57)		(-1.65)		(-1.77)
... × **1**(second half)		-0.020			-0.001		-0.032		-0.002
		(-0.69)			(-0.08)		(-0.82)		(-0.12)
chdist(PS)				-0.100***					
				(-6.83)					
... × **1**(second half)				-0.005					
				(-0.34)					
Constant	0.875***	0.799***	0.754***	0.391***	0.440***	0.791***	0.649***	0.864***	0.779***
	(39.30)	(28.32)	(57.19)	(17.64)	(20.91)	(56.57)	(16.69)	(33.78)	(15.19)
Individual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
P(pooling)	0.300	0.127	0.042	0.000	0.001	0.148	0.260	0.120	0.021
N	3276	3276	3366	3366	3366	3162	3162	3072	3072
Adj. R^2	0.160	0.173	0.103	0.301	0.304	0.219	0.259	0.243	0.259

t statistics in parentheses. Standard errors are clustered at the session level.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

B Instructions

Welcome. This is an experiment on individual decision making in groups, and you will be paid for your participation in cash at the end of the experiment. The entire experiment will take place through computer terminals, and all interactions between participants will take place through the computers. You will remain anonymous to me and to all the other participants during the entire experiment; the only person who will know your identity is the Lab Manager who is responsible for paying you in the end. Moreover, it is important that you do not talk or in any way try to communicate with other participants during the experiment.

We will start with a brief instruction period. During the instruction period, you will be given a complete description of the experiment and will be shown how to use the computers. You must take a quiz after the instruction period, so it is important that you listen carefully. If you have any questions during the instruction period, raise your hand and your question will be answered so everyone can hear. If any difficulties arise after the experiment has begun, raise your hand, and an experimenter will come and assist you. Please note that you are not being deceived and you will not be deceived: everything I tell you is true.

Your earnings during the experiment are denominated in tokens. Depending on your decisions, you can earn more tokens or lose some tokens. At the end of the experiment, we will count the number of tokens you have earned in all of the matches and you will receive $1.00 for every 4 tokens. You will be paid this amount plus the show-up fee of $5. Different participants may earn different amounts. Everyone will be paid in private and you are under no obligation to tell others how much you earned.

The experiment will consist of 8 matches. In each match, you will be put in a group with 5 other participants in the experiment. Since there are 12 participants in today's session, there will be 2 groups in each match. You are not told the identity of the participants in your group. Your payoff in each match depends only on your decisions, the decisions of the other 5 participants in your group and on chance. What happens in the other group has no effect on your payoff and vice versa. Your decisions are not revealed to participants in the other group.

We will now explain how each match proceeds. At the beginning of the match, the computer randomly assigns each of you to a group consisting of 6 participants. Next, the computer randomly assigns with equal probability a role to each of the participants as "Subject 1", "Subject 2" and so on up to "Subject 6". Then, the match begins.

Each match consists of several turns. At the beginning of each turn, the computer randomly pairs all subjects within each group with one another. We shall call the subject that you are paired with at each turn as your "Current Partner". Once everyone receives a Current Partner, a turn begins.

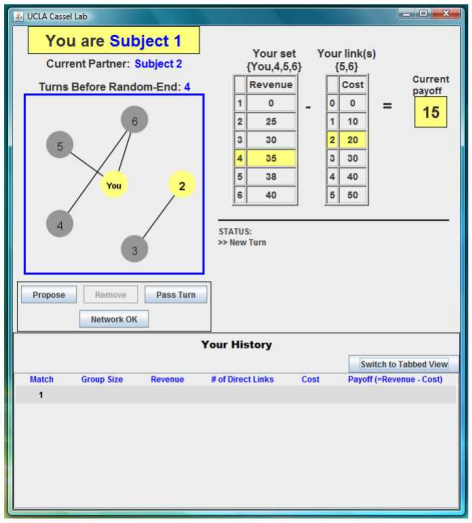


FIGURE 1: BEGIN A TURN

At the beginning of each turn, you will see a screen similar to that shown here. The top panel provides the information and interface that you will use to interact with other subjects within your group. Meanwhile, the bottom panel lists your payoff history throughout the experiment. Payoff information in each match, including the practice matches, is recorded here.

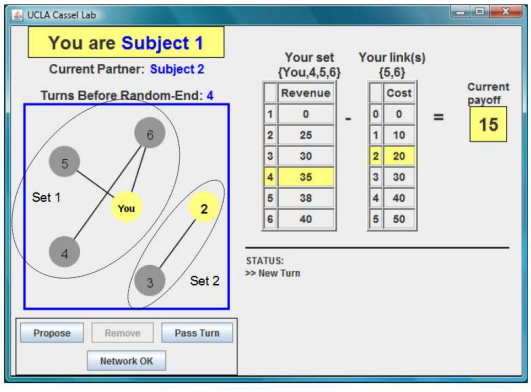


FIGURE 2: SETS, LINKS AND ACTIONS [example 1]

This is the top panel. On the top-left is your role in this match. In this example, you are Subject 1. The computer also informs you of your Current Partner at each turn. In this turn, your Current Partner is Subject 2.

In the middle of the left panel, you will see a network representation of the connections between all subjects in your group. Other subjects in your group are represented by nodes with their role ID numbers. Meanwhile, you are always represented by the center node labeled "YOU". In each turn, the node for your Current Partner is colored YELLOW unlike the rest of the subjects. From the color, you can see here that your Current Partner is Subject 2.

The lines connecting the nodes represent the links between subjects in your group. Everyone in your group sees the same sets of links. In this example, you have direct links to Subjects 5 and 6. Through Subject 6, your are also indirectly connected with Subject 4. Subjects who are either directly or indirectly connected belong in the same "Set". In this example, there are two sets. The first consists of You, Subjects 4, 5, and 6. The second set consists of Subjects 2 and 3.

At each turn, the joint actions of you and your current partner affect how the two of you are linked. You take actions by clicking one of the action buttons below the network representation. Through your actions, you can either propose a link, remove a link, or maintain how you are connected with your partner.

In this first example, since you are not linked to Subject 2, only three actions are available: "Propose", "Pass Turn", and "Network OK". The "Remove" button is not active. Clicking "Propose" lets the computer know that you would like to propose a link with your Current Partner. If your partner does the same, the computer will create a link between you and your partner. Otherwise, no link will be created. In other words, a link is created if and only if BOTH partners propose a link to each other.

If you don't want to link with your Current Partner, you can either click "Pass Turn" or "Network OK". In either case, a link will not be created. However, notice the difference between the two actions. When you pass a turn, you tell the computer that you want to keep the way you are linked with your current partner in this turn. However, you may still want to change how you are linked with some of the other subjects. So, your buttons will remain active in the next turn

Meanwhile, if you choose "Network OK", you tell the computer that as long as the network doesn't change, you are happy with the way you are linked with everyone in your group. Therefore, if you click "Network OK", you won't need to take further actions until the network changes. Your

buttons will therefore be inactive. However, these buttons will immediately become active once the decisions of other pairs either break or make a link. If all active subjects choose "Network OK" in the same turn, then the match ends.

The turn ends once everyone in your group has taken an action. The computer then begins a new turn, and you will be randomly assigned a new Current Partner. Please note that since pairs are selected randomly, you may be paired with the same partner in consecutive turns.

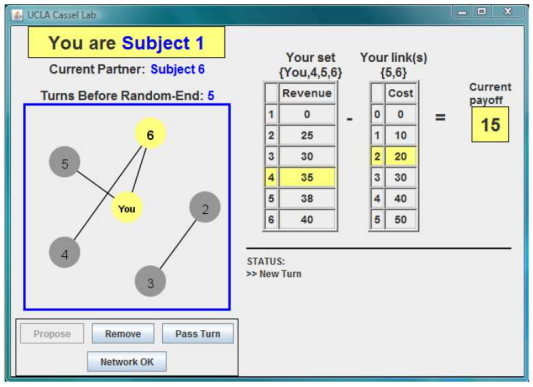


FIGURE 3: SETS, LINKS AND ACTIONS [example 2]

This figure illustrates a new turn in which you are paired with Subject 6. Now, since you are already directly linked with this subject, the three actions available to you are: "Remove", "Pass Turn" and "Network OK". The "Propose" button is deactivated in this turn.

Your link with Subject 6 will remain intact only if BOTH you and Subject 6 don't want to remove it. If at least one subject in the pair wants to remove it, your direct link with your Current Partner will be broken at the end of the the turn. Obviously, the link will also be broken if both subjects in a pair choose to remove it.

In each match, the computer will continue to generate new turns for at least 12 turns unless all subjects choose "Network OK". However, if a match does not end after 12 turns, the match enters the random-end stage. In the random-end stage, at each turn, the computer randomly decides whether it will end the match or generate a new turn. Each time, there is a 20% probability that it will decide to end the match. On average, this implies about 5 additional turns in each match. The number of remaining turns

before this random-end stage is displayed above the network representation.

The network representation updates links that are made and broken in real time. You can see changes to the network immediately after each pair makes their decisions within each turn. Similarly, you can also keep track of changes within each turn through the "Status" indicator on the lower right panel. This status indicator resets at each new turn.

We will next discuss about the payoff. Your payoff depends on the size of your set and the number of direct links at the end of the match. Your set size, which is the number of subjects who are either directly or indirectly connected to you, determines your revenue. Meanwhile, your cost is determined by the number of direct links you have.

The right panel provides you with all of the information necessary to calculate your payoff. The table on the left gives you the revenue schedule for different set sizes. Above it, you can see the list of subjects in your set. In this example, your set consists of You and Subjects 4, 5, and 6. Therefore, as part of a set of size 4, your revenue is 35.

Next to the revenue table is the cost schedule for different numbers of direct links. Each direct link incurs a constant cost. In this particular example, the cost for each link is 10 and, therefore, the total cost is 10 times the number of subjects with whom you are directly linked. Above that table, you can see that you are directly linked to Subjects 5 and 6. Since you have two direct links, the current total cost is 20 tokens.

Your current revenue and cost at any stage of the game are highlighted in YELLOW. They are updated in real time as the actions of subjects make and break links within each turn. The rightmost box entitled "Current Payoff" calculates your payoff at each stage of the game. The current payoff is simply the revenue minus cost, which in this case is 15. This payoff information is also updated in real time. Note that the revenue and cost tables may change from match to match.

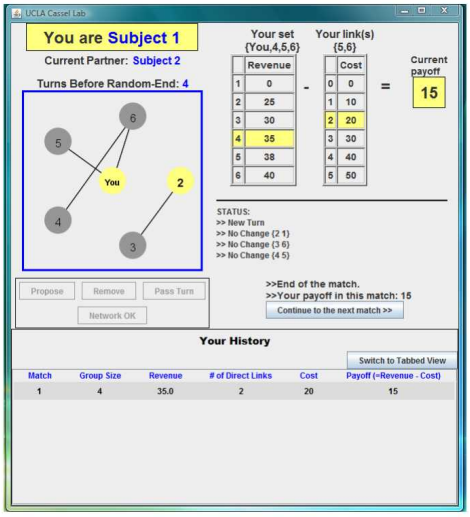


FIGURE 4: END OF A MATCH

This figure illustrates what you will see at the end of a match. Below the status indicator, you will see your payoff for this match. At the end of the match, please click "Continue to the Next Match". In each new match, you will be randomly assigned to a new group. A new match will begin only after all groups have completed their matches. This continues for 8 matches, after which the experiment ends.

At the end of the final match in the experiment, you will see the following screen.

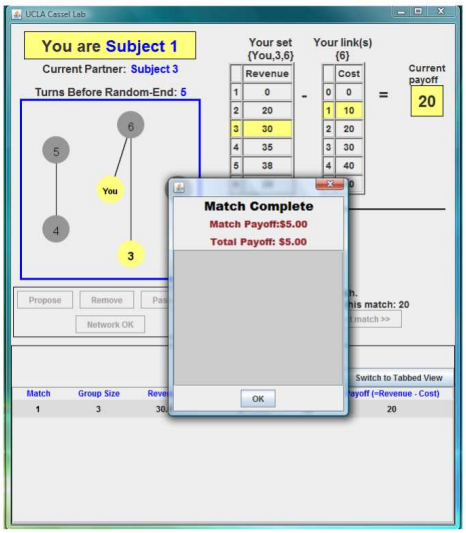


FIGURE 5: END OF A SESSION

This final screen tells you the total payoff that you will receive for this experiment. When you see this screen, don't click OK until you have written down your total payoff on the payoff sheet provided. After you have written down your total payoff, click OK to conclude the session. (*)

The following slides summarize the rules of the experiment:

Summary of the rules (1/2)

- This session consists of 8 matches. In each match, you will be put with 5 other participants in a group. Groups are reshuffled in each new match.
- Each match consists of several **turns**.
  - In each turn, you are randomly paired with a current partner in your group
  - A turn ends once everyone in your group has taken an action.
- Use one of four **action buttons** to affect your link with your partner:
  - **Propose** tells the computer you want to propose a link with a partner. A link is created if and only if **both** you and your partner propose.
  - **Remove** tells the computer you want to remove an existing link with a partner. A link is removed if **either** you or your partner wants to remove it.
  - **Pass turn** tells the computer you want to keep the state of your connection with your partner.
  - **Network OK** tells the computer you are happy with the current network configuration. After choosing "Network OK", you don't need to take subsequent actions until the network changes.

Summary of the rules (2/2)

- A match ends in one of the following two ways:
  - After all active participants have chosen "Network OK" in the same turn;

    OR
  - If the match has not ended after 12 turns, the computer randomly decides the end of the match. Any extra turn afterward may be the last with a 20% chance.
- Your payoff is revenue minus cost. The revenue depends on the size of your set, while the cost depends on the number of your direct links.
- A new match begins only after all groups have finished their matches. In a new match, you will be randomly matched with another 5 participants. Moreover, the revenue and cost tables may also change between matches.

We will now begin the Practice session and go through two practice matches to familiarize you with the computer interface and the procedures. During these practice matches, please do not hit any keys until you are asked to. Remember, you are not paid for these matches. At the end of the practice matches you will have to answer some review questions.

Throughout the session, pay attention to the network representation display and status indicators. Also, notice the movements of the yellow highlights on your Revenue and Cost tables, as well as updates to your Current Payoff.

[START GAME]

You have just received a new turn. First, pay attention to your role. If you are Subject 1, 2, or 3, please click "Propose". For Subject 1, 2, or 3, notice a link has just been created between you and your partner if your partner is also Subject 1, 2 or 3.

Now, if you are Subject 4, 5, or 6, please click the "Pass Turn" button. Notice here that a link is created if and only if BOTH partners propose a link. If only one partner proposes a link, no link is created.

You have moved to a new turn. We will now see how the "Network OK" action works. If you are either Subject 5 or 6, please click "Network OK". For the rest of the group, please click "Pass Turn".

You have moved to a new turn. For Subjects 5 or 6, since the network has not changed after you clicked "Network OK", all of your buttons are now inactive. Notice that they will become active following a change in the network.

For others, please check your Current Partner. If your partner is not Subject 5 or 6, click the "Remove" button if it's active or "Propose" otherwise. For Subjects 5 and 6, notice how a change in the network activates your buttons.

If you are not Subject 5 or 6 and your buttons are still active, please click "Pass Turn". If you are Subject 5 or 6 and your buttons are active, please click "Pass Turn". Notice here that if your buttons are inactive due to a "Network OK" action in a previous turn, a change in the network will immediately activate your buttons. In the following, we will do the same exercise for Subjects 1 to 4.

You have moved to a new turn. If you are Subject 3 or 4, please click "Network OK". For the rest of the group, please click "Pass Turn".

You have moved to a new turn. Subjects 3 and 4, notice that your buttons are inactive. If the network changes in this turn, your buttons will become activated.

For all others, check your Current Partner. If your partner is not Subject 3 or 4, click "Remove" if it's active or click "Propose" otherwise. If you are not Subject 3 or 4 and your buttons are still active, click "Pass Turn". Now, if you are Subject 3 or 4, please click "Pass Turn".

You have moved to a new turn. If you are either Subject number 1 or 2, please click "Network OK". For the rest, please click "Pass Turn".

You have moved to a new turn. For Subject 1 or 2, your buttons are now inactive. For all others, if your Current Partner is not Subject 1 or 2, click the "Remove" button if it's active, or click "Propose" otherwise. For everyone else who has not taken an action, please click "Pass Turn".

You have moved to a new turn. Notice from the message above the network display that this is the last turn before the random-end stage. During the paid match, you will have 12 turns before entering this stage. If the match has not ended after 12 turns, the computer will randomly decide the end of the match.

We will now deliberately end the match. If your buttons are active, please click the "Network OK" button. This ends the first practice match. The bottom part of your screen contains a table summarizing the results for all matches you have participated in. This is called the history screen. It will be filled out as the experiment proceeds. Now click "Continue to the Next Match". We will now begin with the second practice match.

[NEXT MATCH]

You are in a new match. Note here that the revenue and cost tables have changed as they may during the real matches. We'll now examine the behavior of the "Remove" action.

If you are either Subject 2, 4, or 6, please click "Remove". For Subjects 1, 3, and 5, please click "Pass Turn". Hence, notice that a link is broken if at least one of the partners chooses to remove it.

You have moved to a new turn. Next, we'll see what will happen if the network changes within the turn in which you click "Network OK". If you are Subject number 1, 3, or 5, please click the "Network OK" button. For all others, please click your "Remove" button.

You have moved to a new turn. For Subjects 1, 3, or 5 notice that if in the previous turn the network changed after you clicked "Network OK", your action buttons are active in this turn. If the network did not change after you clicked "Network OK", your buttons remain inactive. Now, if you are either Subject 2, 4, or 6, click "Network OK". For all others, if you haven't taken an action in this turn, please click the "Remove" button if it's active, or "Propose" otherwise.

You have moved to a new turn. Similarly for Subjects 2, 4, and 6, notice that if in the previous turn the network changed after you clicked "Network OK", your buttons are now active. If the network did not change after you clicked "Network OK", your buttons are still inactive. If the network changes in the same turn and after you choose "Network OK", your buttons stay active in the following turn.

We will now end the match. If your buttons are active, please click "Network OK". This ends the second practice match.

*** END OF PRACTICE SESSION ***

The practice matches are over. Please click "Continue to the next match" and complete the quiz. It has 8 questions in two pages. You will move to the next page once everyone in your group has completed the questions in that page correctly. On your table, you will find the screenshots that you will need to answer these questions. Raise your hand if you have any questions.

[WAIT for everyone to finish the quiz]

Are there any questions before we begin with the paid session? We will now begin with the 8 paid matches. Please pull out your dividers. If there are any problems or questions from this point on, raise your hand and an experimenter will come and assist you.

[START MATCH 1]

[After MATCH 8, read:]

This was the last match of the experiment. Now, please write down your ID on the payment sheet. Your ID is located on top of your physical monitor and it began with CASSEL. At this point, if you haven't clicked "Continue to the next match", please do so. Your total payoff is displayed

on your screen. Please record this payoff in the earned column of your sheet and sign it. Once you have written it down, please click OK.

Your Total Payoff will be this amount rounded up to the nearest dollar plus the show-up fee of $5. We will pay each of you in private in the next room. Remember you are under no obligation to reveal your earnings to the other subjects.

If you are done, please line up behind the yellow line until the lab manager calls you to be paid. Do not converse with the other subjects or use your cell phone. Thank you for your cooperation.